

Week 3 — Spread & distributions

Business Forecasting · LIVE pilot (slides + exercises in one deck)

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Where we are

C1 Framing → C2 Describing data → C3 Uncertainty → C4–C5 Regression → C6 Forecasting

We are wrapping up **C2 — describing data**: this week we cover both the **spread** of the data and the **shape** of its distribution.

- **Spread**: how dispersed is the data — **range, IQR, variance, standard deviation**, all summarized in a **boxplot**.
- **Distributions**: how the data is spread out — **histograms** and distribution **shape** (modality, skewness).

The exercises you used to open in a separate learnr window appear **as slides** — the code runs right here in the browser.

Population, parameters & statistics

Parameters vs Statistics

- You need to know how many people in Mexico have diabetes

Parameter

- Call μ_d the proportion of Mexican population which has diabetes
 - Usually the parameter is an **unknown** number **describing the whole population**
 - You want to learn what it is
 - In our example, μ_d is a parameter that you want to learn
 - More generally, parameter describes an aspect of the entire population

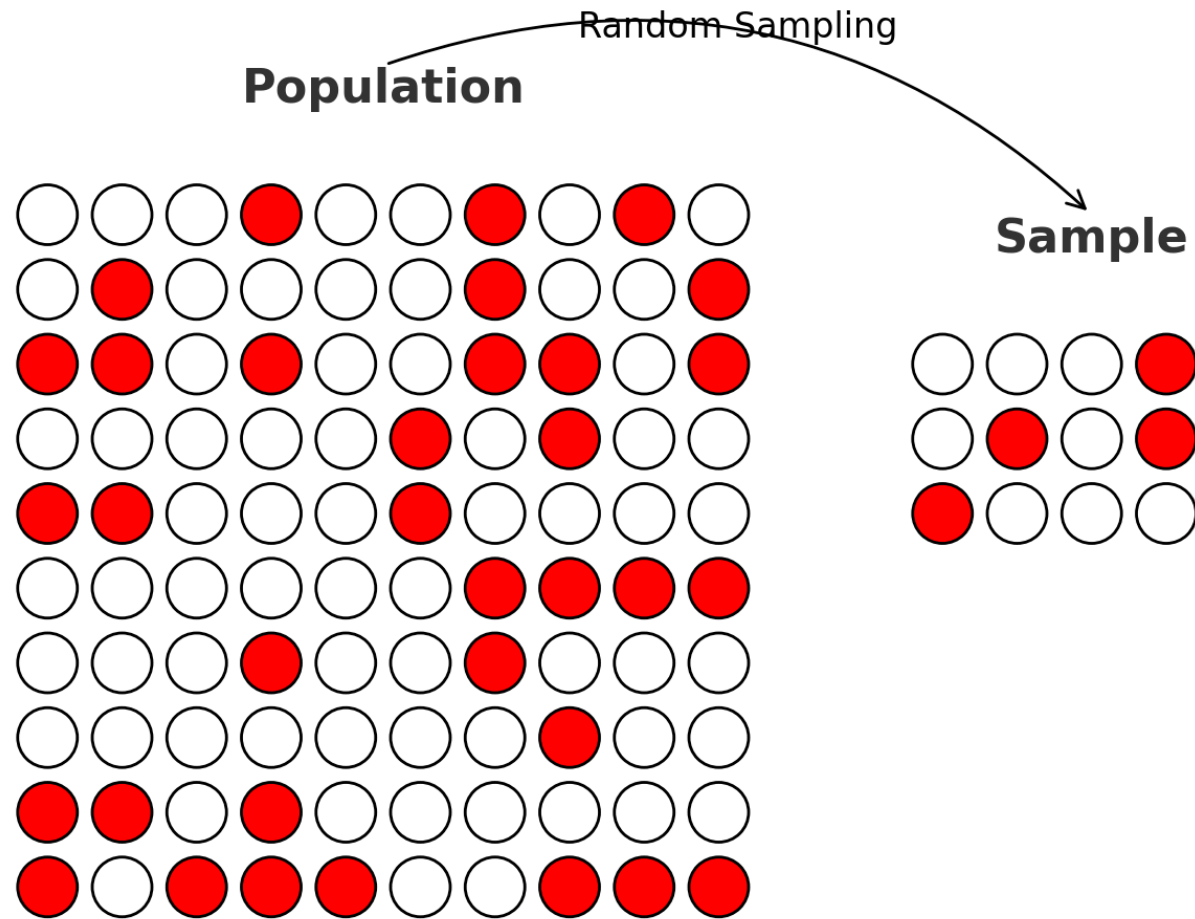
Parameters vs Statistics (cont.)

- You need to know how many people in Mexico have diabetes

Statistic

- But you don't have data on the whole population. At best you can get a sample from a survey
 - So you will try to estimate this parameter with sample
 - Statistic is a **guess of the parameter** which can be **calculated from the sample**
 - You will calculate a statistic $\hat{\mu}_p$ which is the proportion of diabetics in the sample

Sampling: categorical variable

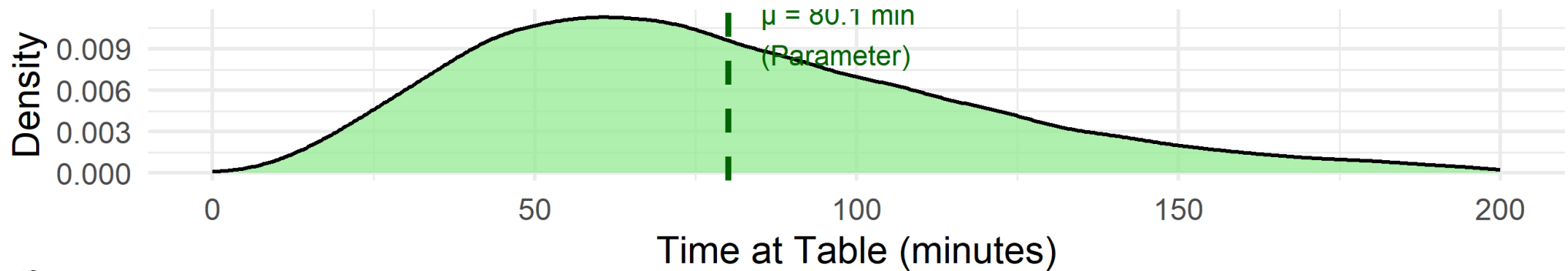


Parameter (Population Diabetes %): 34.00% Statistic (Sample Diabetes %): 33.33%

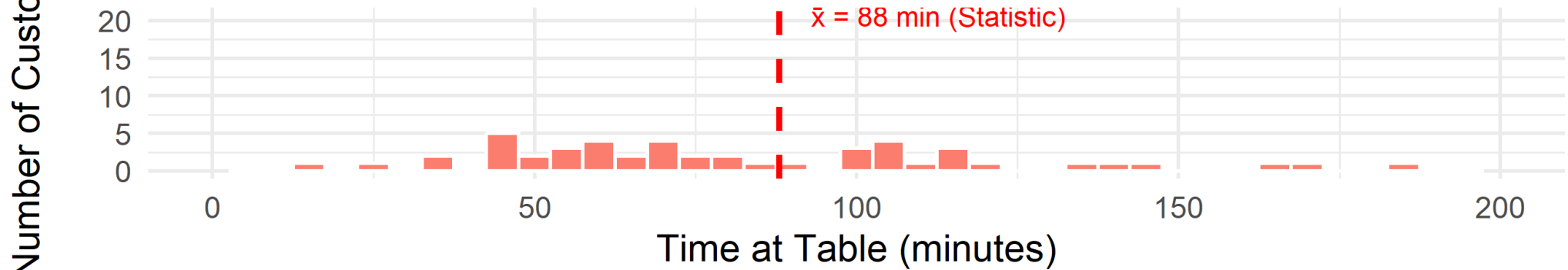
Sampling: continuous variable

Customer Time at Table: Population vs. Sample Mean

Population: All Customers (1 Month)



Sample: 50 Customers



Parameters vs Statistics: examples

- What is population, sample, parameter and statistic in the following examples?
- You want to know the probability that a user who got a match on tinder will go out on a date with that person. You survey 1000 users and ask them about each match they got if they went on a date. You then calculate the share of dates which ended up in a match for these users.
- You want to know what whether starbucks baristas are faster than Cielito Querido baristas. You go to 10 starbucks and 10 Cielito Querido and measure the time it takes to make a coffee. You then calculate the average time it takes to make a coffee in each of these chains.
- You want to know the average age of people who go to the gym. You go to a gym and ask 100 people about their age. You then calculate the average age of these people.
- You want to know the variance of internet speed during in Mexico City. You visit 500 households and calculate the variance of their internet speed.

Parameters vs Statistics: looking ahead

- What are some examples of the statistics we will use and why do we care?

Measures of Dispersion

Why dispersion matters

- Suppose a store has an average daily revenue of 10 000 pesos
- It could be that on each day it has exactly 10 000 pesos revenue
- Or it could be that on half of days it gets 20 000 pesos but on other half it gets 0 pesos
- Dispersion makes a big difference, especially when trying to understand risk!

Range

- **Range** the difference between minimum and maximum value in the data

$$R = x_{max} - x_{min}$$

- How far apart are the oldest and the youngest person **with** diabetes?
- And is that group more or less spread out than people **without** diabetes?



Exercise 1 — Range

Compute the range of `age` separately for people with diabetes (`diabetes == 1`) and for people without. Report both gaps, then say which group looks more spread out.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Subset first, then measure.
```

Range: what you found

- With diabetes: $R = 77 = 97 - 20$
- Without diabetes: $R = 91 = 111 - 20$
- The non-diabetic group has the wider range — but that number was decided by **two people**: the oldest and the youngest. Every other observation was ignored.

Interquartile Range

- **Interquartile range** is the difference between the first and the third quartile of the data:

$$IQR = q_3 - q_1$$

- What is the IQR of age in people with diabetes?
- And what **share** of that group has an age between q_1 and q_3 ?



Exercise 2 — Interquartile range

For people with diabetes: compute the IQR of `age`, then get q_1 and q_3 yourself and check that their difference is the same number. Finally compute the share of that group whose age lies **between** q_1 and q_3 — decide what it *should* be before you run it.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

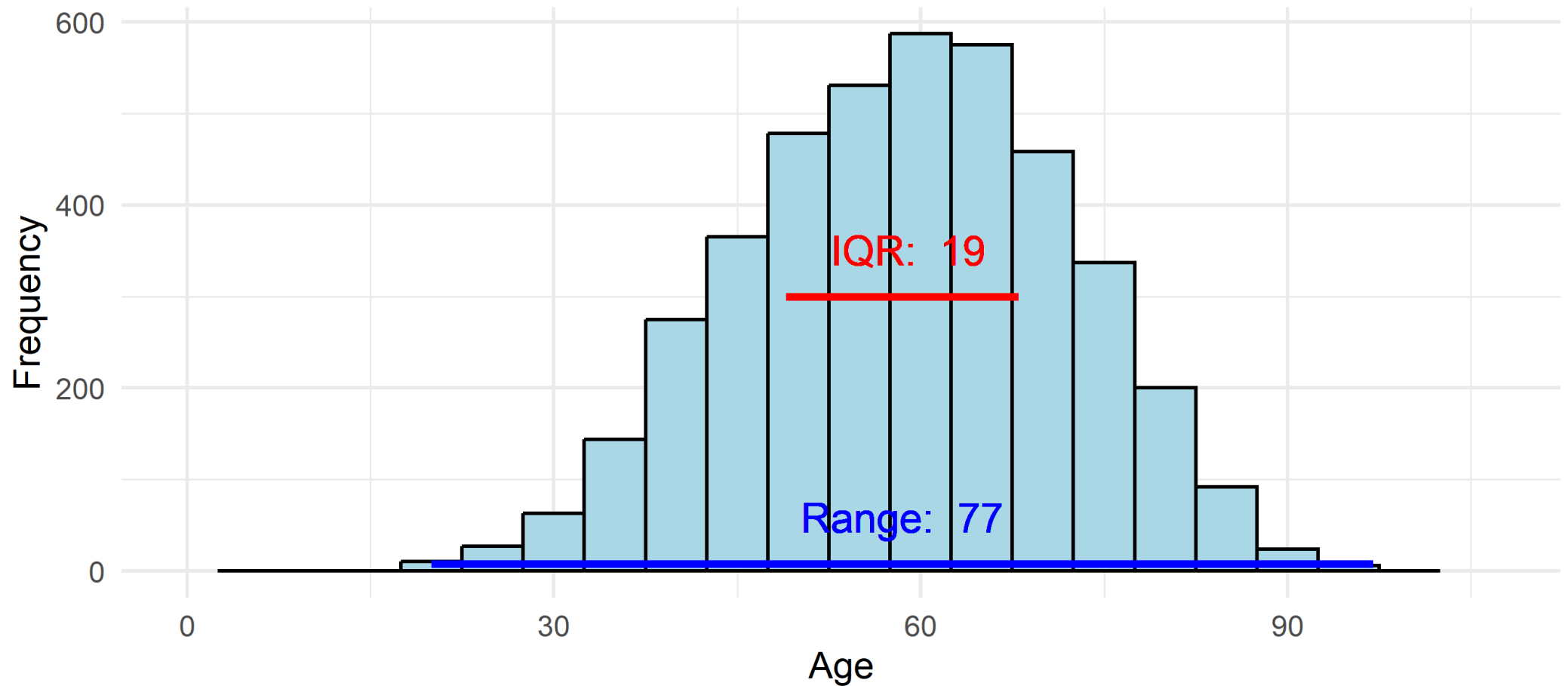
```
1 d1 <- Health_data$age[Health_data$diabetes == 1]
2 # my guess for the share between q1 and q3: ____
```

Interquartile range: what you found

- $IQR = 19 = 68 - 49$
- 50% of the sample is between q_1 and q_3 — that is what the quartiles mean.
- The share you computed is 0.533, not exactly 0.5: age is recorded in whole years, so many people sit **exactly on** q_1 or q_3 and get counted in.

Interquartile Range (cont.)

- 50% of the sample (age of people with diabetes) is between q_1 and q_3



One typo, four statistics

A nurse types a patient's weight as **700 kg** instead of 70.

Which of these does that single keystroke move — the **mean**, the **median**, the **range**, the **IQR**?



Exercise 11 — Break it

Take the first 50 weights, corrupt one value, and recompute everything. Report which statistics barely moved and which one exploded.

Exercise

↺ Start Over

💡 Show Hint

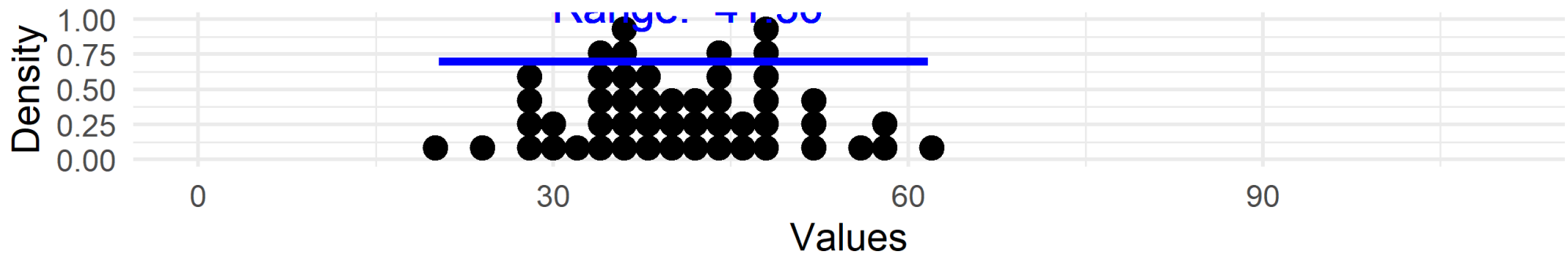
▶ Run Code

```
1 w <- Health_data$weight[1:50]
2 c(mean = mean(w), median = median(w), range = diff(range(w)), IQR = IQR(w))
3
4 # Now break it: set the first weight to 700 and recompute the same four numbers.
```

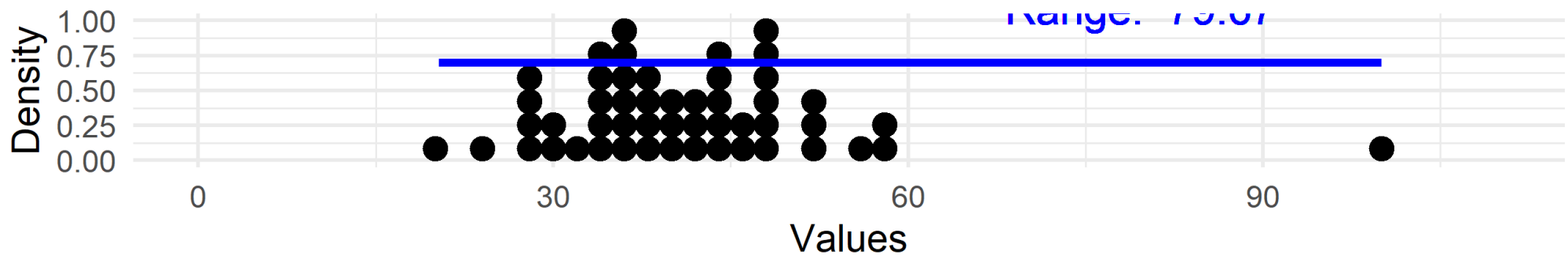
Range: sensitive to outliers

- Same 50 observations in both panels — panel B just moves one point out to 100.
- The range is **very sensitive to outliers**: it is built from the two extremes.

A



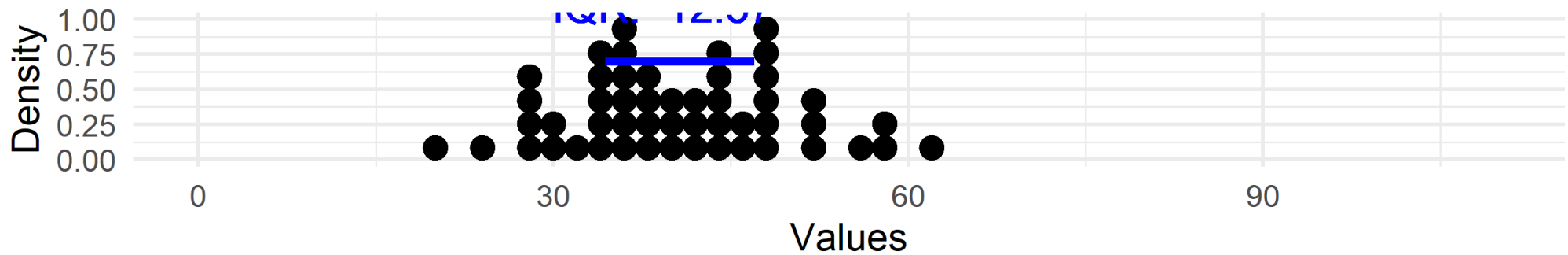
B



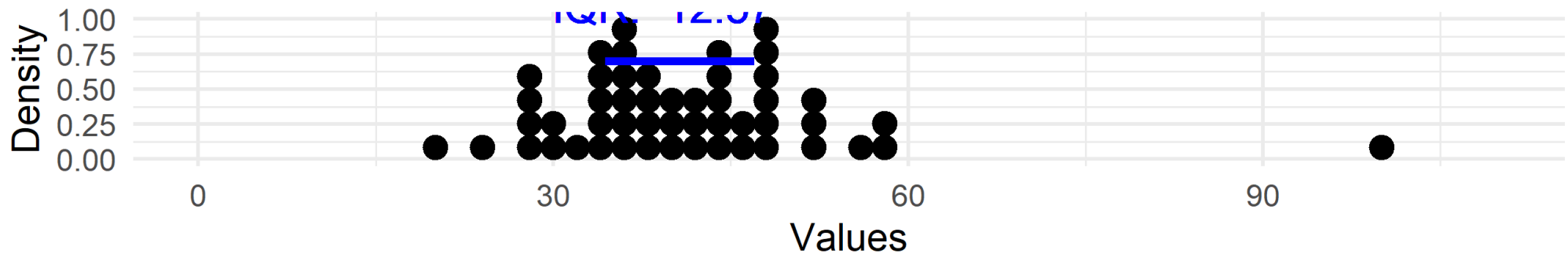
IQR: robust to outliers

- Same two panels, now with the IQR drawn instead of the range.
- The IQR is **robust**: the box hardly notices the moved point.

A



B



Example with data

- What is the IQR?

Video title	Views
TikTok Video 1	30
TikTok Video 2	17
TikTok Video 3	22
TikTok Video 4	24
TikTok Video 5	23
TikTok Video 6	19

First 6 of 20 videos shown.

Example with data (cont.)

Here is a (smaller) data on distribution of how many views have various tik-tok videos.

- Suppose that all views triples and 1000 additional people viewed them as well

$$y_i = 3x_i + 1000$$

- What is new IQR?

Video title	Old views	New views
TikTok Video 1	30	1090
TikTok Video 2	17	1051
TikTok Video 3	22	1066
TikTok Video 4	24	1072
TikTok Video 5	23	1069
TikTok Video 6	19	1057



Exercise 14 — Predict, then check

The 20 view counts are in `views`. Compute q_1 , q_3 and the IQR. Write down your prediction for what each becomes under $y_i = 3x_i + 1000$ — then build `y` and check.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 views
2 quantile(views, c(0.25, 0.75))
3 IQR(views)
4
5 # my prediction for the new IQR: ____
6 # now build y and find out
```

IQR under a linear transformation

- Order of observations was not affected, so same observations correspond to the first and the third quartile

$$q_1^{New} = 3q_1^{Old} + 1000$$

$$q_3^{New} = 3q_3^{Old} + 1000$$

- And more generally, for

$$y_i = bx_i + a$$

and $b > 0$

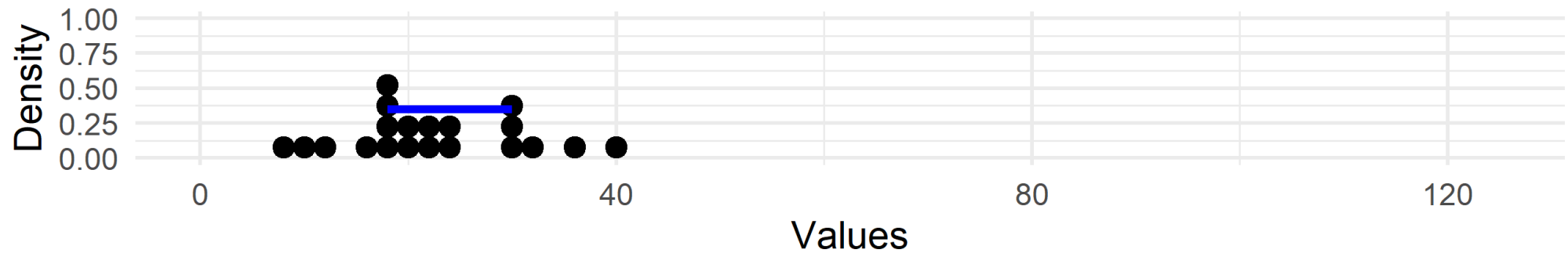
$$v_p^y = bv_p^x + a$$

- if $b < 0$ then the order reverses.
- So what does it mean for IQR?

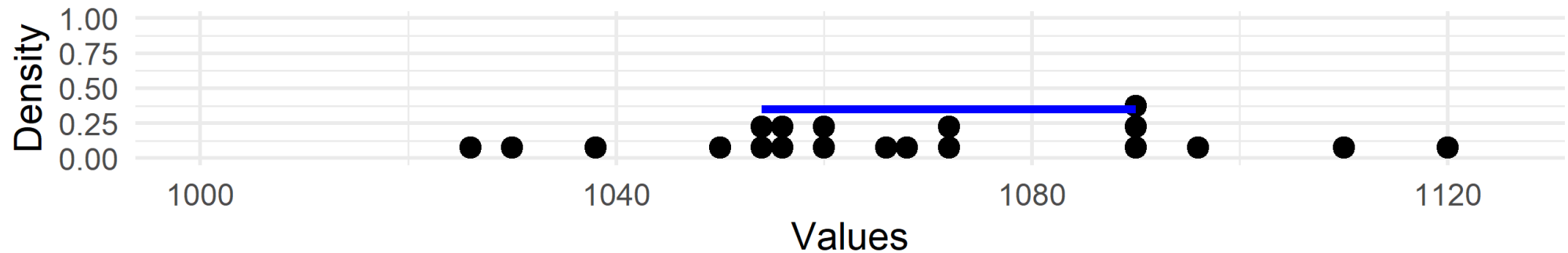
$$IQR^{New} = q_3^{New} - q_1^{New} = 3q_3^{Old} - 3q_1^{Old} = 3 * IQR^{Old}$$

IQR after the transformation

A



B



Variance & Standard Deviation

Variance measures how far an average observation is from the mean:

- **Population variance:**

$$\sigma^2 = E[(X - \mu)^2] = E[X^2] - \mu^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2 = \frac{1}{N} \left(\sum_{i=1}^N x_i^2 - N\mu^2 \right)$$

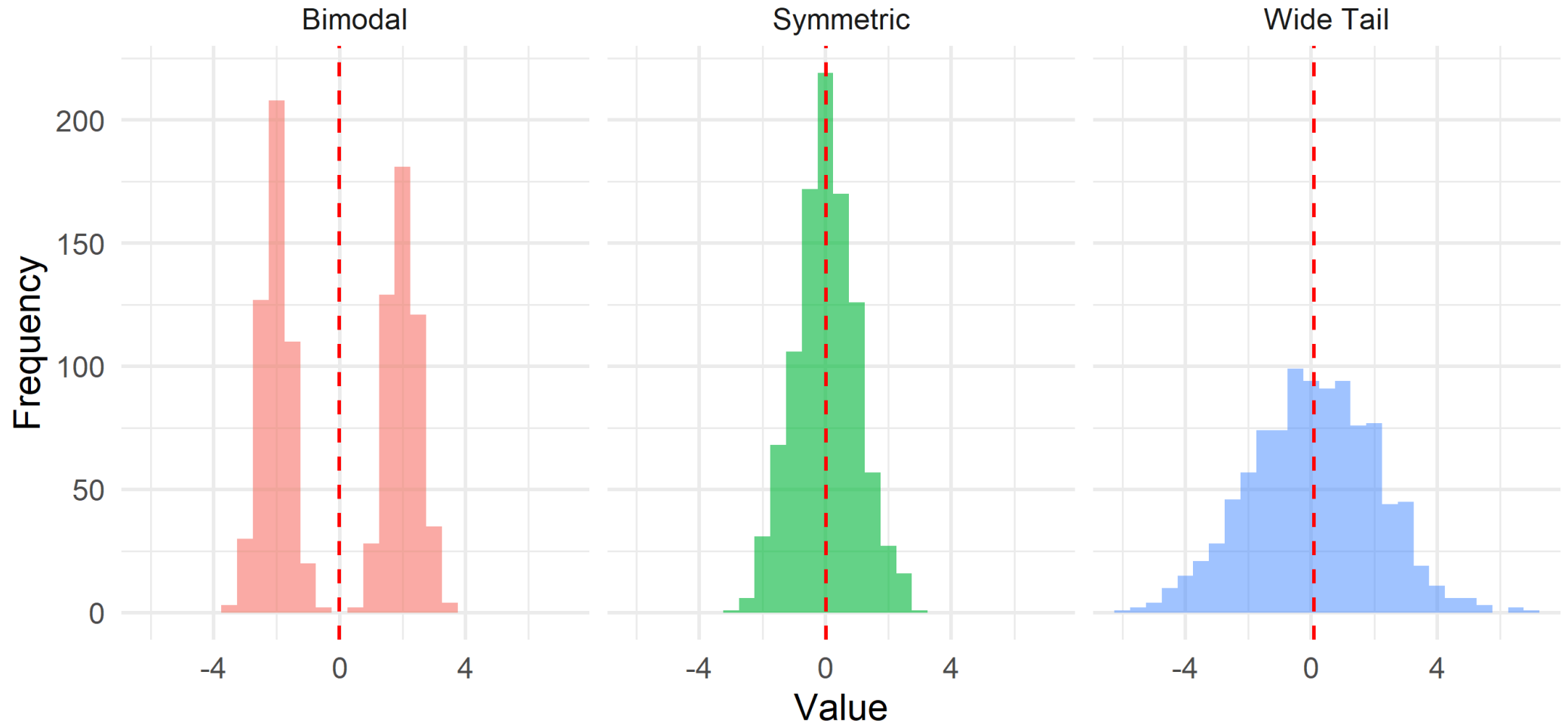
For Discrete Variables it can be: $\sigma^2 = \sum_k P(X = k)(k - \mu)^2$, where k is any possible value that X can take.

- Variance does not have the right units since it squares everything, so we take the square root:
- **Population standard deviation:**

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$

Same center — but rank them by spread

All three have a mean of ~0 (dashed line). Rank them from least to most spread out.





Exercise 3 — Rank them, then compute

The three samples from the previous slide are rebuilt for you. Commit to a ranking first, then compute the variance and the standard deviation of each and see whether your eye was right.

Exercise

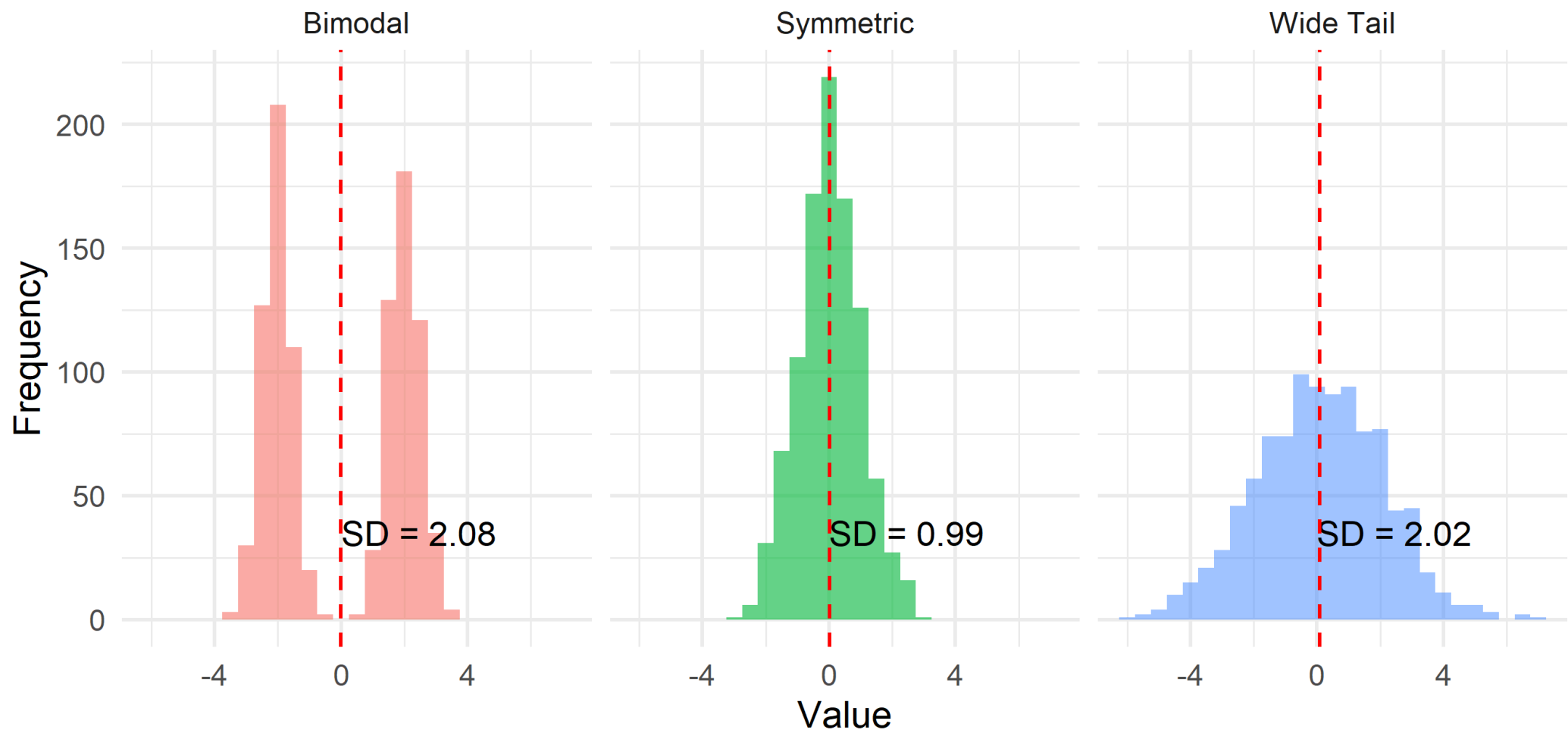
↺ Start Over

💡 Show Hint

▶ Run Code

```
1 set.seed(123)
2 symmetric <- rnorm(1000, mean = 0, sd = 1)
3 wide_tail <- rnorm(1000, mean = 0, sd = 2)
4 bimodal   <- c(rnorm(500, mean = -2, sd = 0.5), rnorm(500, mean = 2, sd = 0.5))
5
6 # my ranking, least to most spread: ____ , ____ , ____
```

Variance: same center, different spread



A manager asks: “spread of what, exactly?”

You hand your manager the variance of **weight**. She asks: “*and that’s in kilograms, right?*”

Is it? And what would happen to that number if the hospital recorded weights in **grams** instead?



Exercise 12 — What are the units?

Compute the variance and the standard deviation of `weight` in kilograms. Then express every weight in **grams** and compute both again. Report the factor by which each one grew — and which of the two you could label “kg” on a chart axis.

Then break it once more: set one weight to 700 kg and recompute the SD. Does the SD behave like the range or like the IQR?

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 w_kg <- Health_data$weight
2 w_g  <- w_kg * 1000
```

Variance & Standard Deviation: two bets

Consider two bets/situations:

- Bet A: with 75% you get 200 pesos and with 25% you lose me 200 pesos
 - 75% chance your life goes normal and you keep making money
 - 25% change your house burns down
- Bet B: with 75% you get 110 pesos and with 25% you get 70
 - When your life goes normal you get 110 (you pay 90 for insurance)
 - When your house burns down you are paid some compensation (70)
- Compute expected value and variance of each bet — which one would you prefer?
- What if I change Bet B to: with 75% you get 109 and with 25% you get 69?



Exercise 19 — Price the two bets

Write the discrete formulas in R — $E[X] = \sum_k P(X = k) k$ and $\sigma^2 = \sum_k P(X = k)(k - \mu)^2$ — and compute the expected value and the variance of each bet. Then do it for the modified Bet B (109 / 69).

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 p <- c(0.75, 0.25)
2 A <- c(200, -200)
3 B <- c(110, 70)
4
5 # E[X] and Var(X) for each
```

The two bets: what you found

- Both A and B have the **same expected value, 100** — the mean cannot tell them apart.
- Variance: A = **30 000**, B = **300**. B is a hundred times less risky for the same average payoff, which is why you would buy it.
- The modified Bet B has expected value **99** and the *same* variance, 300. You give up 1 peso of expected value to keep the low risk — and that 1 peso is the insurer's profit.
- **Dispersion, not the mean, is what you are paying to remove.**

Sample equivalents

- Sample variance:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{n-1} \left(\sum_{i=1}^n x_i^2 - n\bar{x}^2 \right)$$

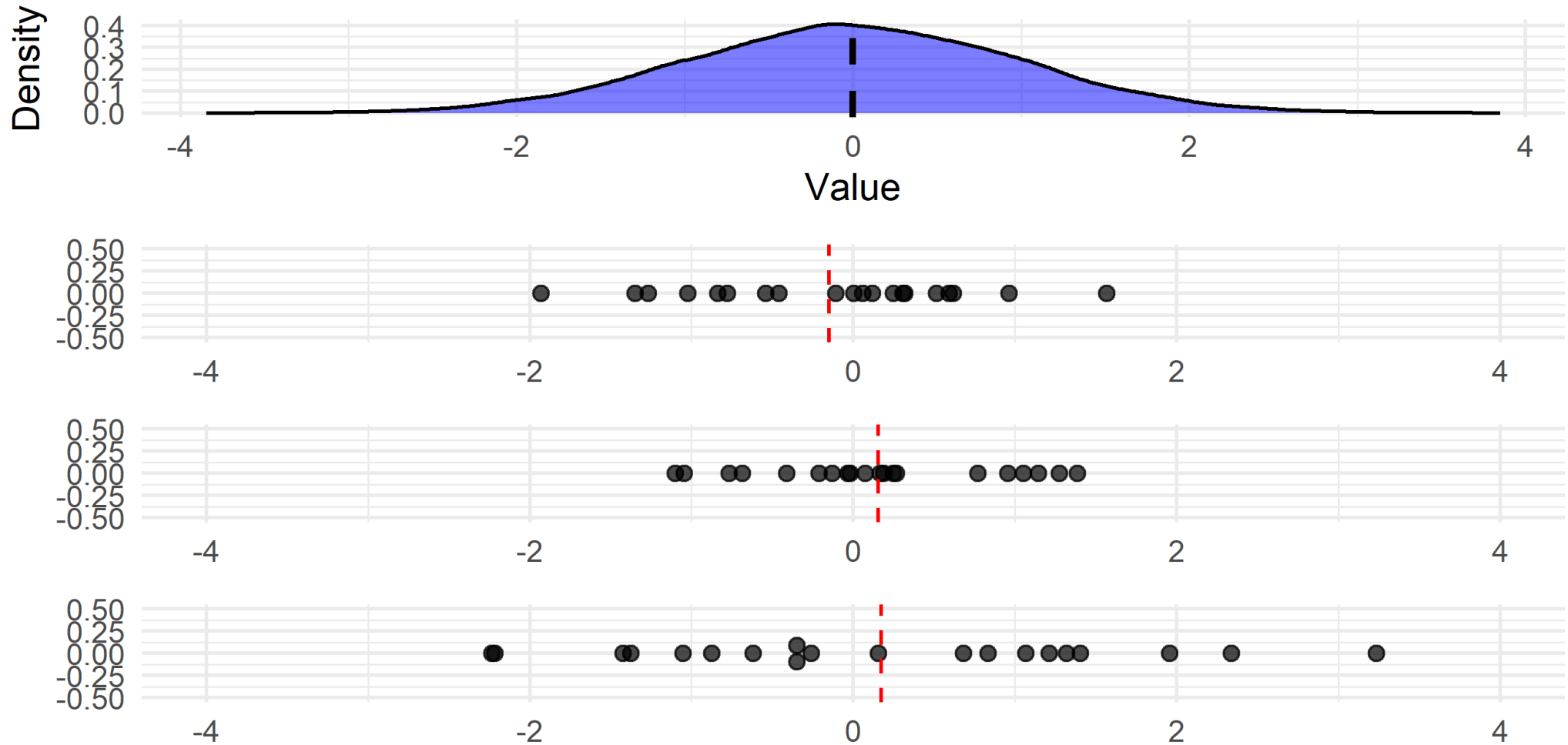
- Sample standard deviation:

$$s = \sqrt{\frac{1}{n-1} \left(\sum_{i=1}^n x_i^2 - n\bar{x}^2 \right)}$$

- Why we divide by $n - 1$ rather than n ?

Why $n-1$: small samples

- **Intuition** - observed values usually fall closer to the sample mean than to the population mean. Distances are artificially small.

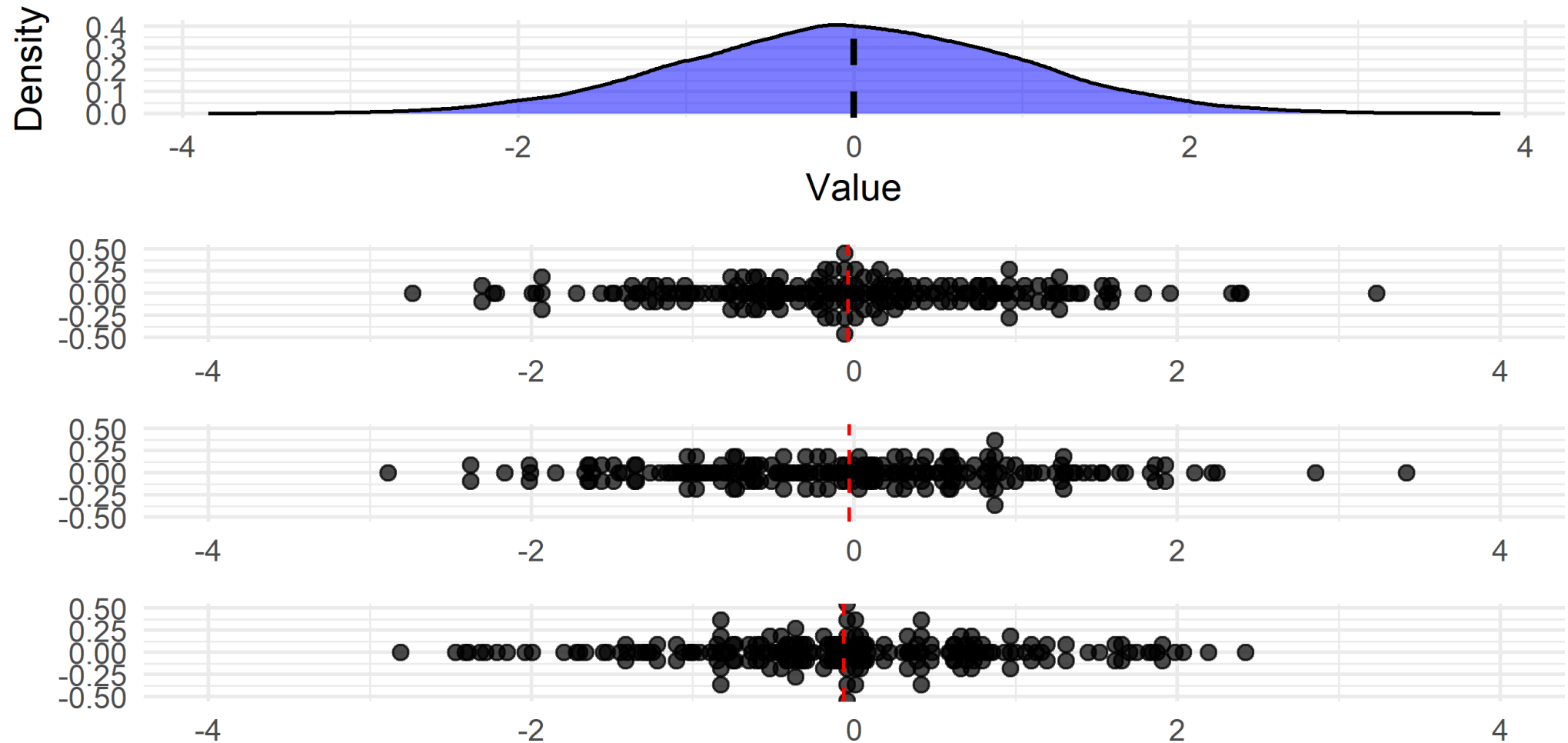


Sample equivalents (cont.)

- So the deviations from the sample mean underestimate the population standard deviation
- So we divide by a smaller number to correct for it
- In big sample $\frac{1}{n}$ and $\frac{1}{n-1}$ are similar, so correction doesn't matter as much

Why $n-1$: big samples

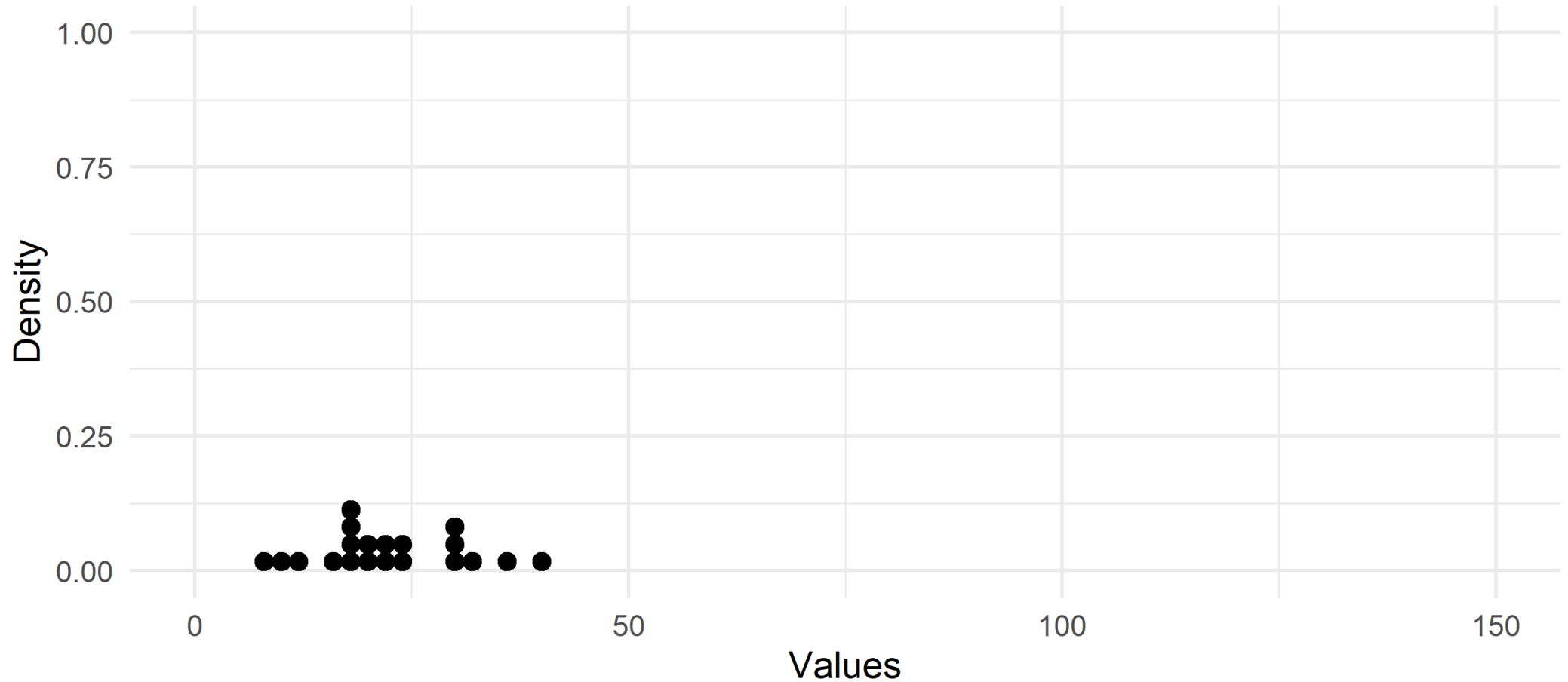
- **Intuition** - in big samples, our estimate of the population mean is already good, no need to correct



Standard Deviation

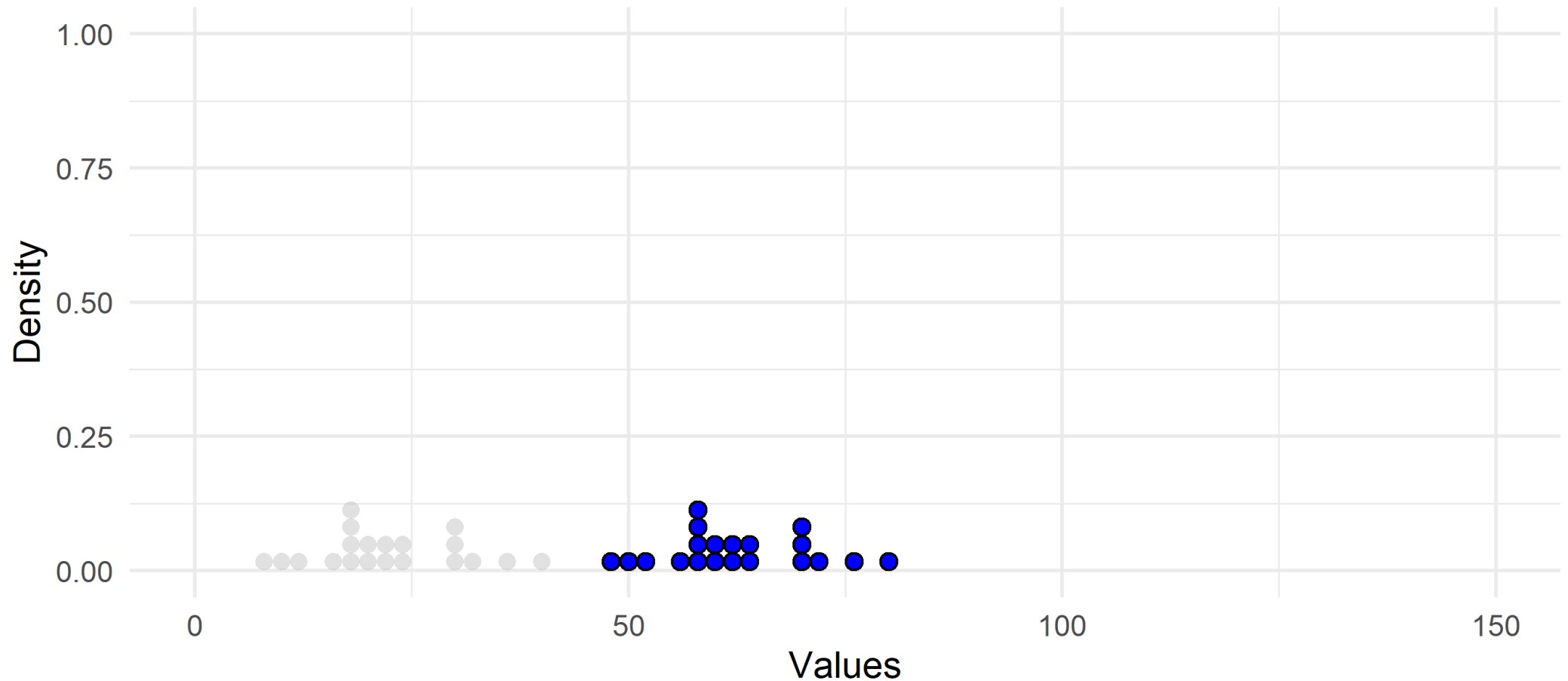
Consider a random variable X with $E(X) = \mu_x$ and standard deviation σ_x .

Ex: X is number of instagram followers distributed like this:



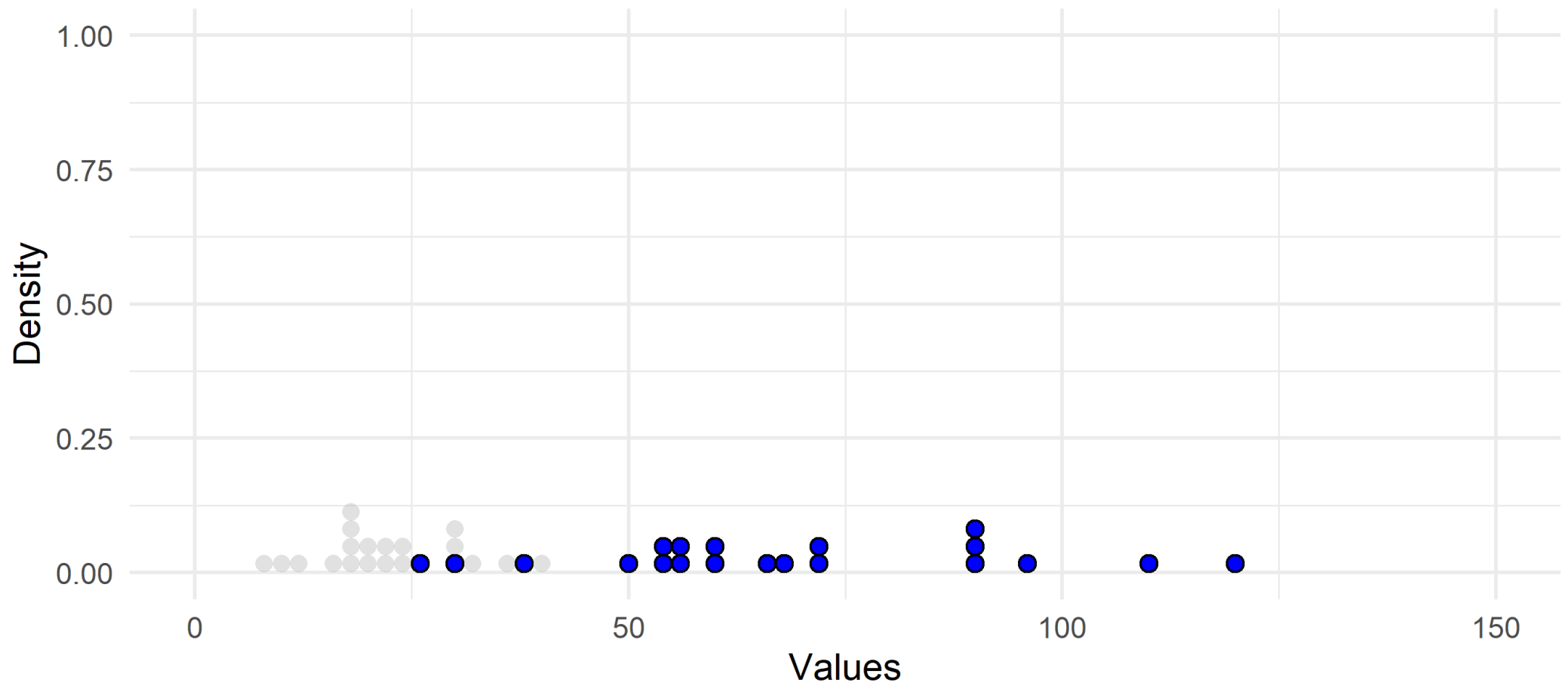
Standard Deviation: add a constant

- What happens to mean and standard deviation if everyone gets 40 more followers?
- $Y = X + 40$



Standard Deviation: multiply by a constant

- What happens to mean and standard deviation if everyone followers get multiplied by 3? (without addition)
- $Y = 3 * X$





Exercise 16 — Shift it, stretch it

Two questions from the last two slides, answered with numbers. Write your prediction first, then compute.

1. Everyone gains 40 followers: $Y = X + 40$. What happens to the mean? To the SD?
2. Everyone's followers triple: $Y = 3X$. What happens to the mean? To the SD? To the **variance**?

Exercise [↺ Start Over](#) [💡 Show Hint](#)

[▶ Run Code](#)

```
1 x <- views
2 c(mean = mean(x), sd = sd(x), var = var(x))
3
4 # my prediction for  $X + 40$  : mean _____ , sd _____
5 # my prediction for  $3X$  : mean _____ , sd _____ , var _____
```

What a linear change does

- $Y = X + a$: the mean shifts by a , the SD does **not move at all** — every observation and the mean move together, so no distance changes.
- $Y = bX$: the mean is multiplied by b , the SD by $|b|$, and the variance by b^2 .

$$E(X + c) = E(X) + c \quad \text{Var}(X + c) = \text{Var}(X)$$

$$E(cX) = cE(X) \quad \text{Var}(cX) = c^2 \text{Var}(X)$$

The algebra behind these four lines is in the appendix.

Box and Whiskers plot

- Helps to see the **distribution** of the data and the **outliers**
 - Outliers reveal anomalies and potential data-collection errors
 - A whisker extends at most $1.5 \times \text{IQR}$ from the box
 - Any point beyond that is flagged as an **outlier**
 - If no point is beyond $1.5 \times \text{IQR}$, the whisker stops at the last datapoint



Exercise 4 — Build one, then read the outliers off it

Draw a boxplot of `weight` and print its five-number summary. Then answer with numbers, not by eye: **above which weight** does R start drawing dots, and **how many** people are out there?

Exercise

↺ Start Over

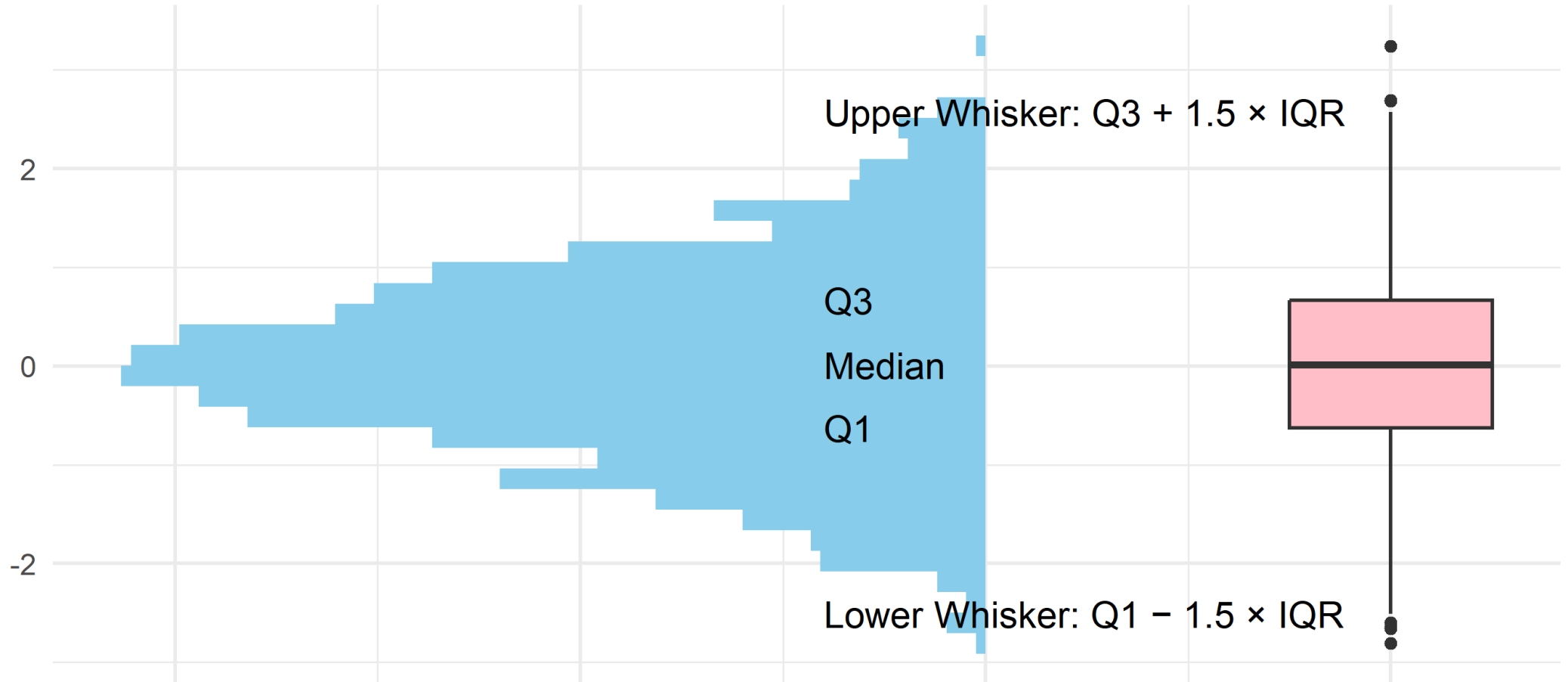
💡 Show Hint

▶ Run Code

```
1 # Your code here
2
3 # Check your cut-off against the rule: q3 + 1.5 * IQR
```

Box and Whiskers plot: anatomy

The box spans Q1–Q3 (the middle 50%); the line inside is the median.



Two questions, one picture

Are men heavier than women in this sample — and is one of the two groups more **variable** than the other?

A boxplot answers both at once. Build it before we show ours.



Exercise 5 — Compare two groups

Draw **weight** by **gender** in a single picture. Read the two answers off it, then confirm them with numbers.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

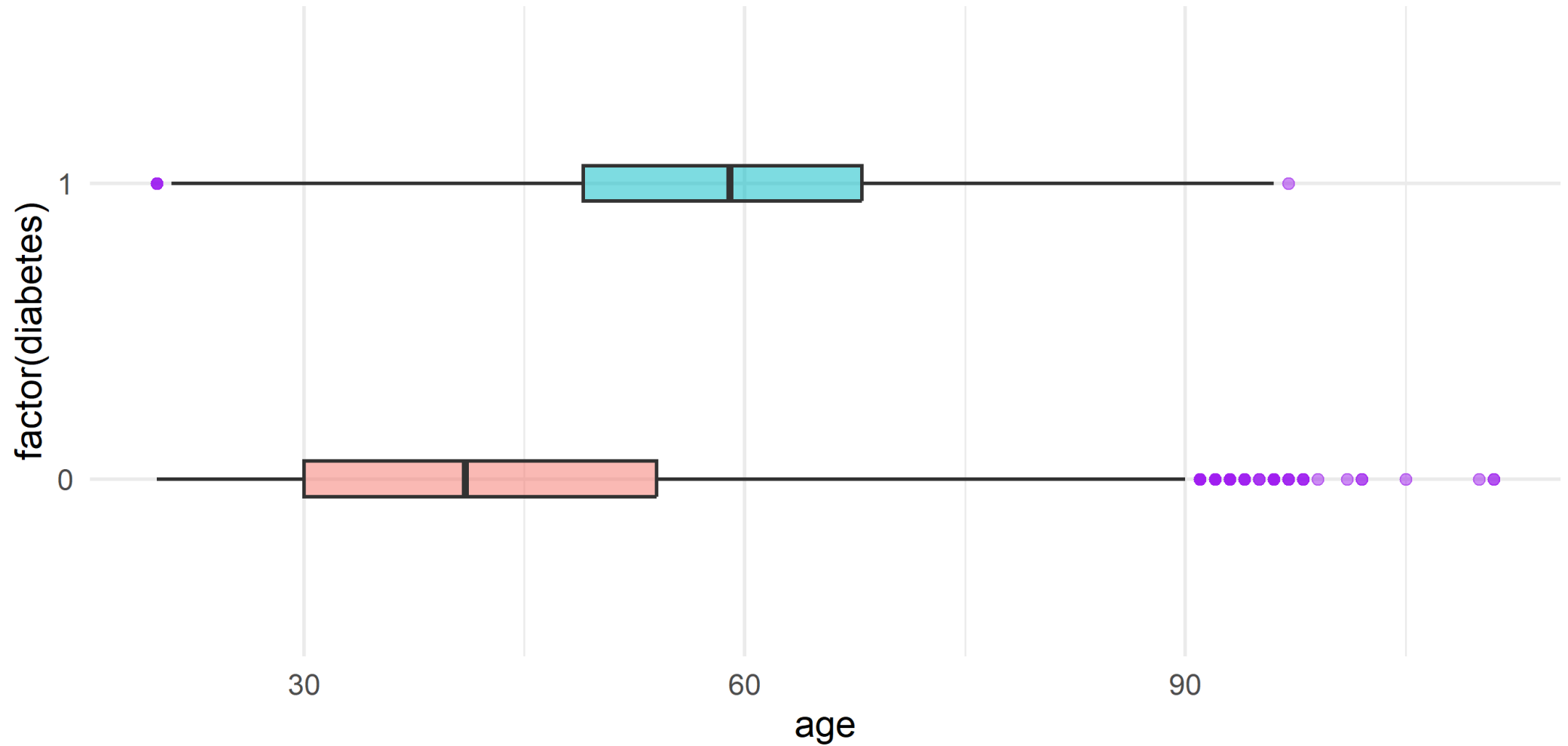
1 # One picture...

2

3 # ...then the numbers behind it.

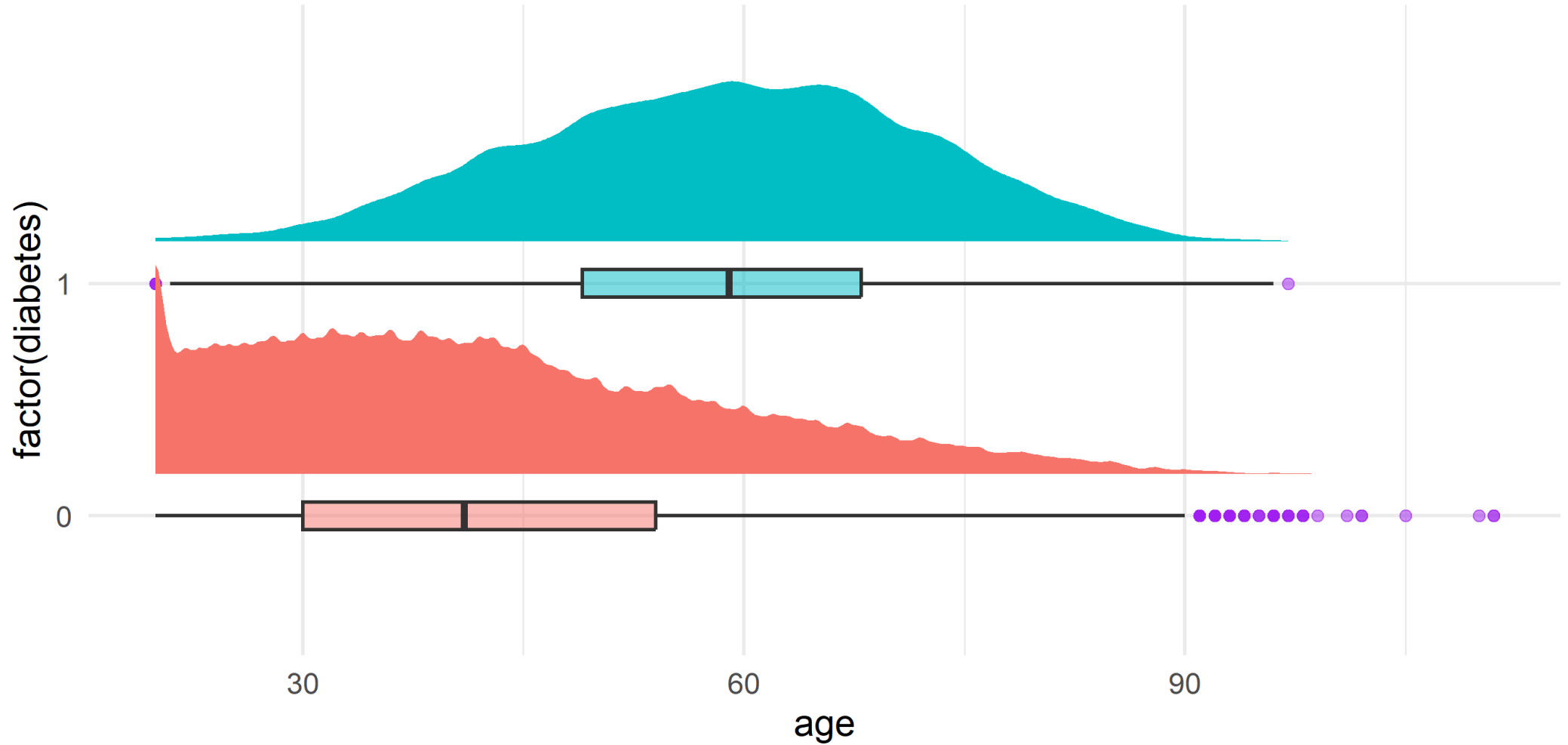
Box and Whiskers plot (cont.)

Dataset comparisons — they summarize data very well (age by diabetes status):



Box and Whiskers plot (cont. 2)

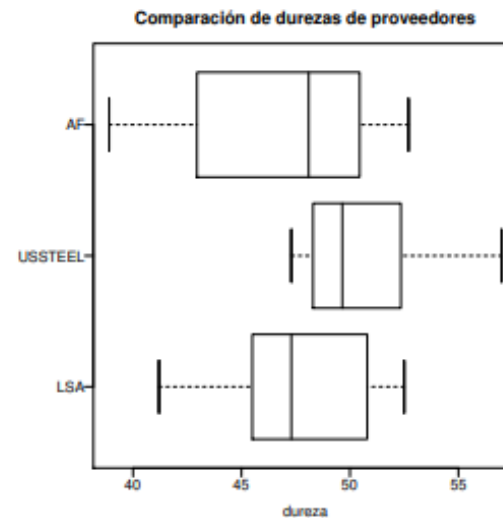
Dataset comparisons — boxplot with the full distribution overlaid (age by diabetes status):



Reading a boxplot (warm-up)

5. [5 puntos] The next plot represents data on the hardness of steel rods for three different suppliers: AF, USSteel, and LSA. Based on this figure, it may be said that:

Hardness of steel rods for three different suppliers



- a) Distributions for all supplier's hardness are skewed to the left.
 - b) AF's hardness seems to have the least dispersion of the three.
 - c) AF's hardness seems to have less dispersion than that of LSA.
 - d) USSTEEL hardness distribution is skewed to the right.
6. [5 puntos] The prior plot representing the hardness of steel rods for three different suppliers, allows us to identify useful:
- a) central tendency and location measures.
 - b) insights about the distribution's dispersion.
 - c) asymmetry among distributions.
 - d) all of the above.

Distributions

How is the data *spread out*? Now picture the *whole* distribution.

The obvious first move

Week 2 gave us the frequency table: count how often each value occurs, then plot the shares. `age` is a number, not a label — but nothing stops us doing it anyway.

Does the same recipe work for **every** numeric variable?



Exercise 17 — Try the recipe on both

Build the relative-frequency plot of `age` the Week-2 way. Then run **exactly the same code** on `weight`. Compare the two pictures and say what broke.

Exercise

↺ Start Over

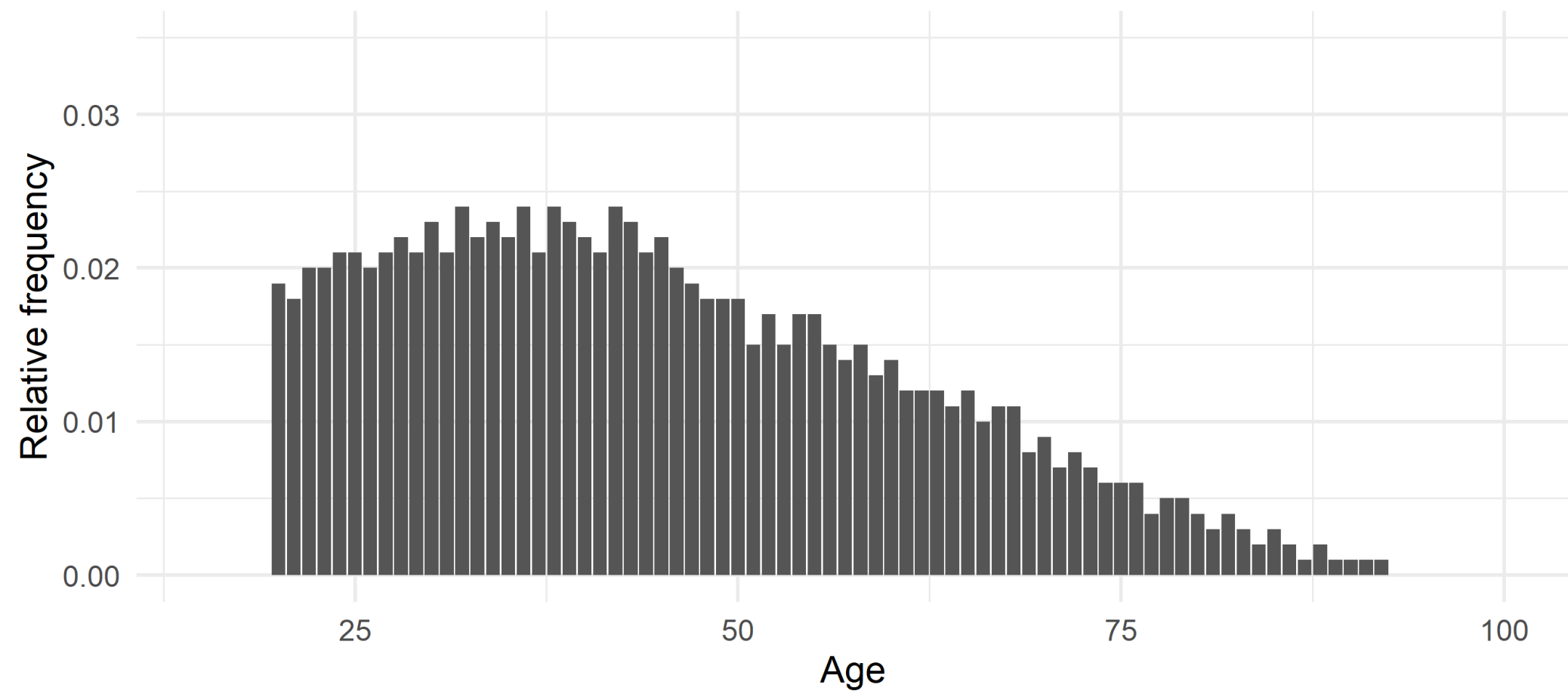
💡 Show Hint

▶ Run Code

```
1 # age first
2 p_age <- prop.table(table(Health_data$age))
3
4 # now the identical recipe on weight
```

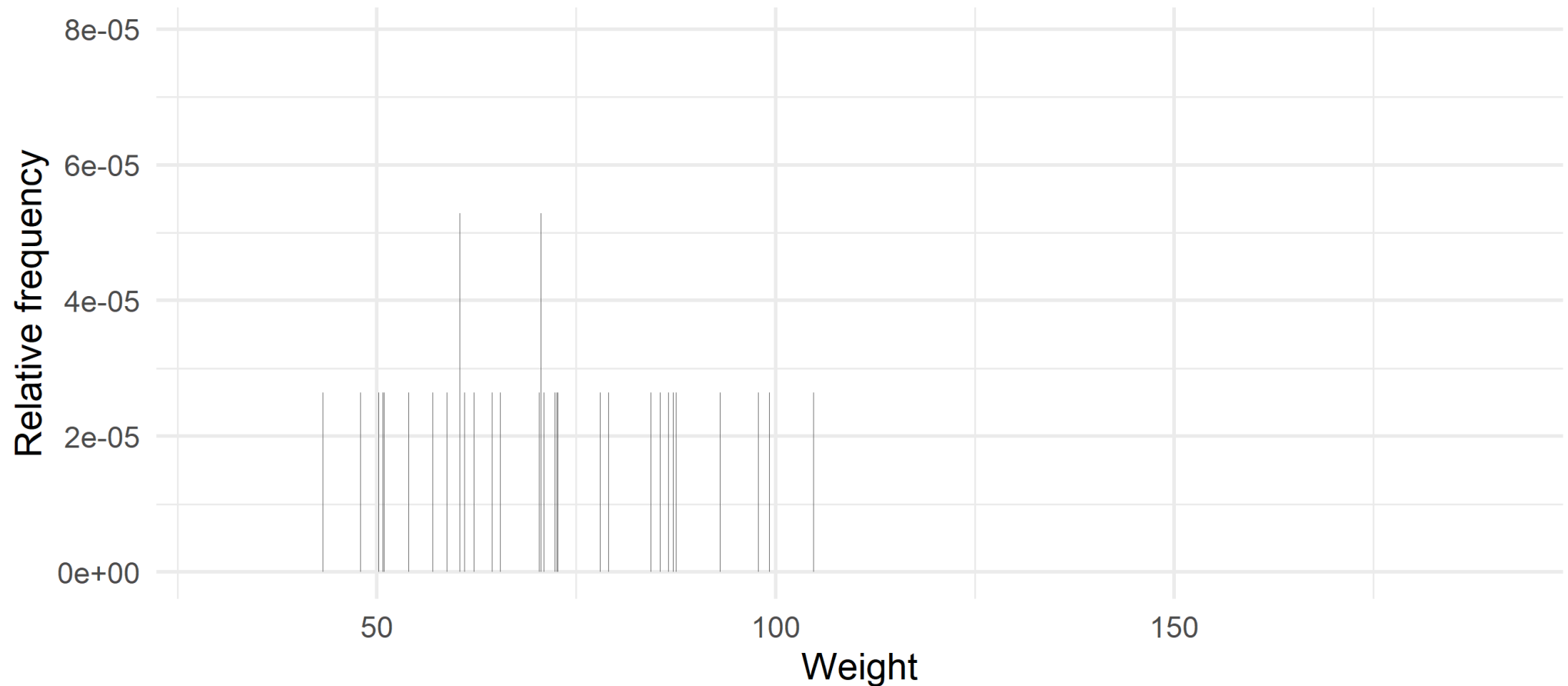
Frequency distribution of a numeric variable

Compare age in the adult population (20+) — how often does each value occur?



Continuous variables: the problem

- The same recipe on a **continuous** variable like weight:



- Most values **never repeat**, so each has a tiny relative frequency — the plot is useless.

Histograms

Solution: Group similar values together

- Construct intervals and show how many observations are in a given interval

Process

1. Decide how many intervals
2. And how wide they are
3. Then calculate the absolute and relative frequencies of each interval
4. Plot it with bars



Exercise 6 — Build the first one

Build a histogram of `weight`. Then answer from **your own picture**, before we show any: roughly where does it peak, and is either tail longer than the other?

Exercise

↺ Start Over

⚠ Show Solution

▶ Run Code

1 # Your code here

2

3 # peak at about ____ kg ; the longer tail is on the ____

Histograms: choosing bin width

- I want k (example $k=5$) equal intervals
- Divide the **range** of the data (max – min) into k equal intervals

```
max_value <- max(Health_data$weight)
min_value <- min(Health_data$weight)
range <- max_value - min_value
```

```
[1] "Range= 190.81 - 30.37 = 160.43"
```

- So how wide is each of the 5 intervals?
- Range $\div 5 \approx 32$ kg per interval; the first starts at the min (30.37), the last ends at the max (190.81).

Same data, how many stories?

The number of bins is **your** choice, not the data's. Does the choice change what a manager would conclude?



Exercise 7 — Break the bins

Draw the same histogram four times, with `breaks = 5, 30, 100, 1000`. Look at all four, then answer: which one hides the skew, which one is unreadable noise, and which one would you put in the report?

Exercise

↺ Start Over

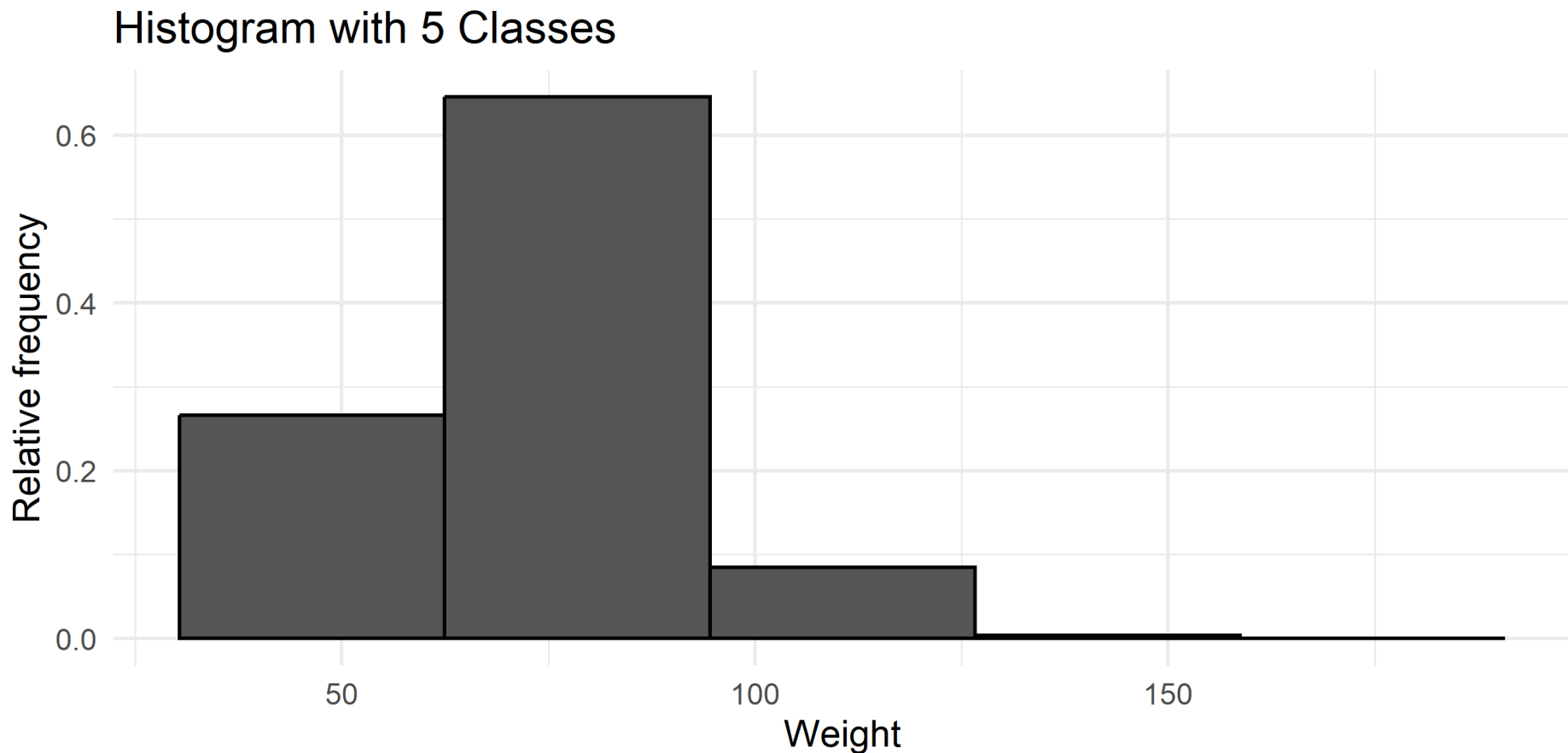
💡 Show Hint

▶ Run Code

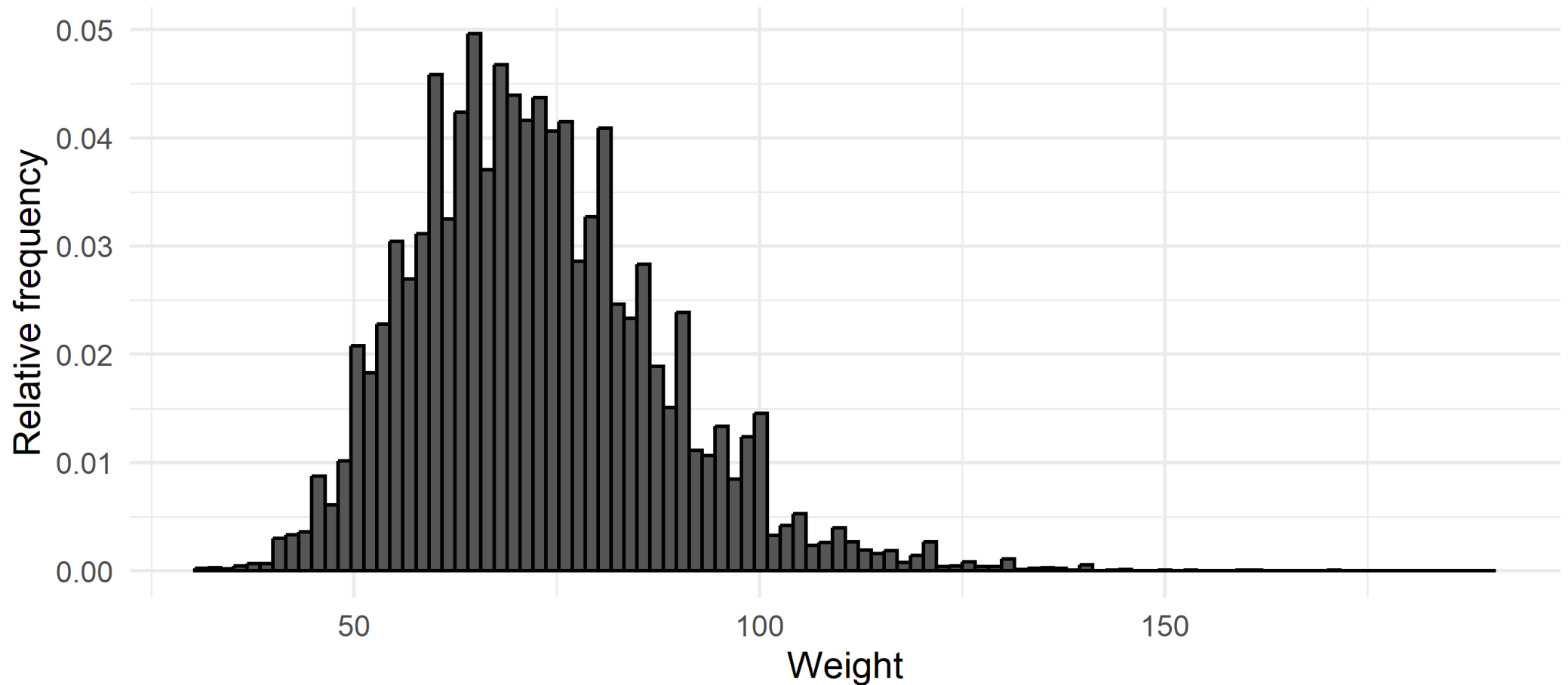
```
1 par(mfrow = c(2, 2)) # four plots in one window
```

Histogram with 5 classes

- Midpoint = center of the interval (center of the bar)
- P_i = cumulative frequency: share of observations in this or a smaller interval
 - So $P_{(62.46-94.55)}$ answers “what share of people weigh less than 94.55 kg?” — the appendix has you build that column yourself.



Histogram with 100 classes



- Helps to see the distribution and outliers — but is more always better?
- The bars change every time you change [breaks](#). Is there anything underneath that **doesn't**?



Exercise 8 — Find the curve the histogram is chasing

Put the histogram on the **density** scale and lay a smooth density curve on top of it. Then check the claim above: rerun it with `breaks = 5` and with `breaks = 200` — does the curve move when the bars do?

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

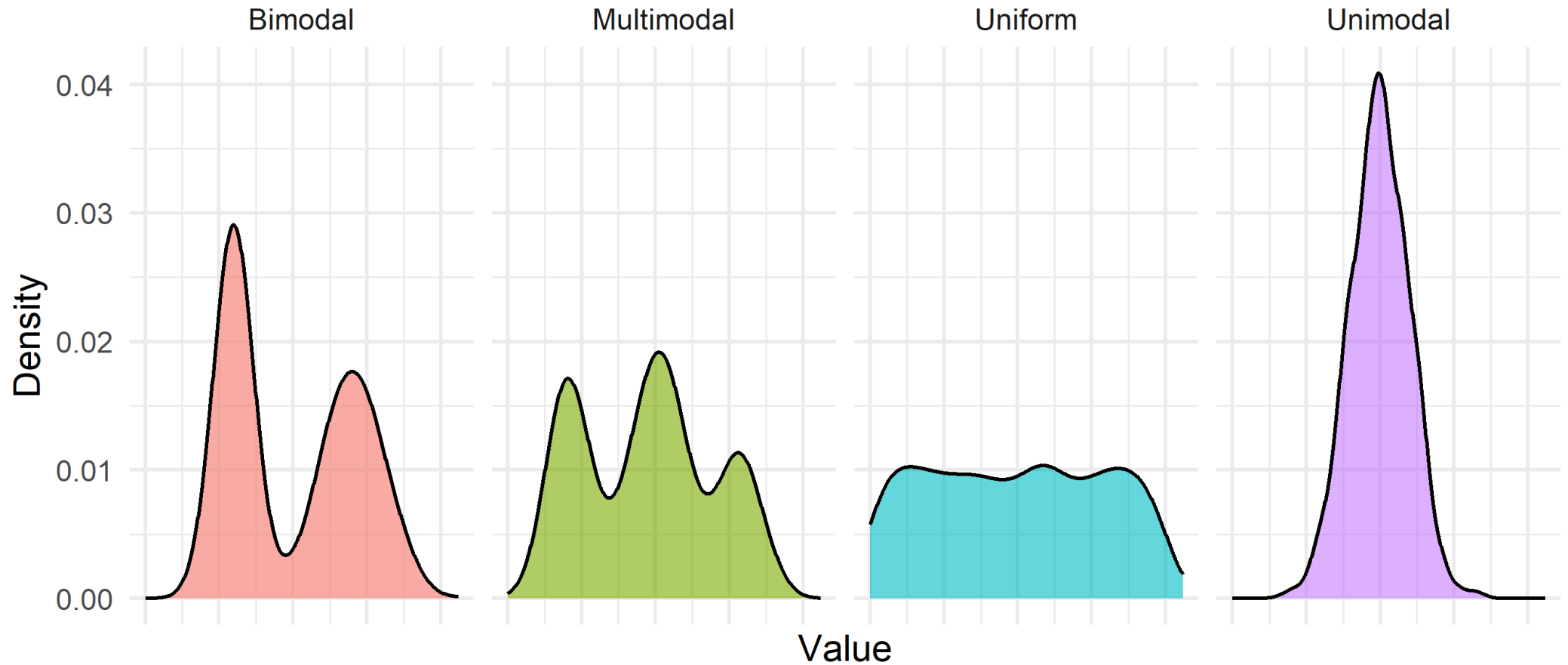
```
1 # Your code here
```

What the bars are approximating

- With smaller and smaller intervals, the histogram tends to the **probability density function** — the curve you just drew.
- The histogram is a *sample* picture of it; the PDF is the population object.
- The **PDF and CDF** sections in the appendix take this apart properly.

Distribution Shapes: Modality

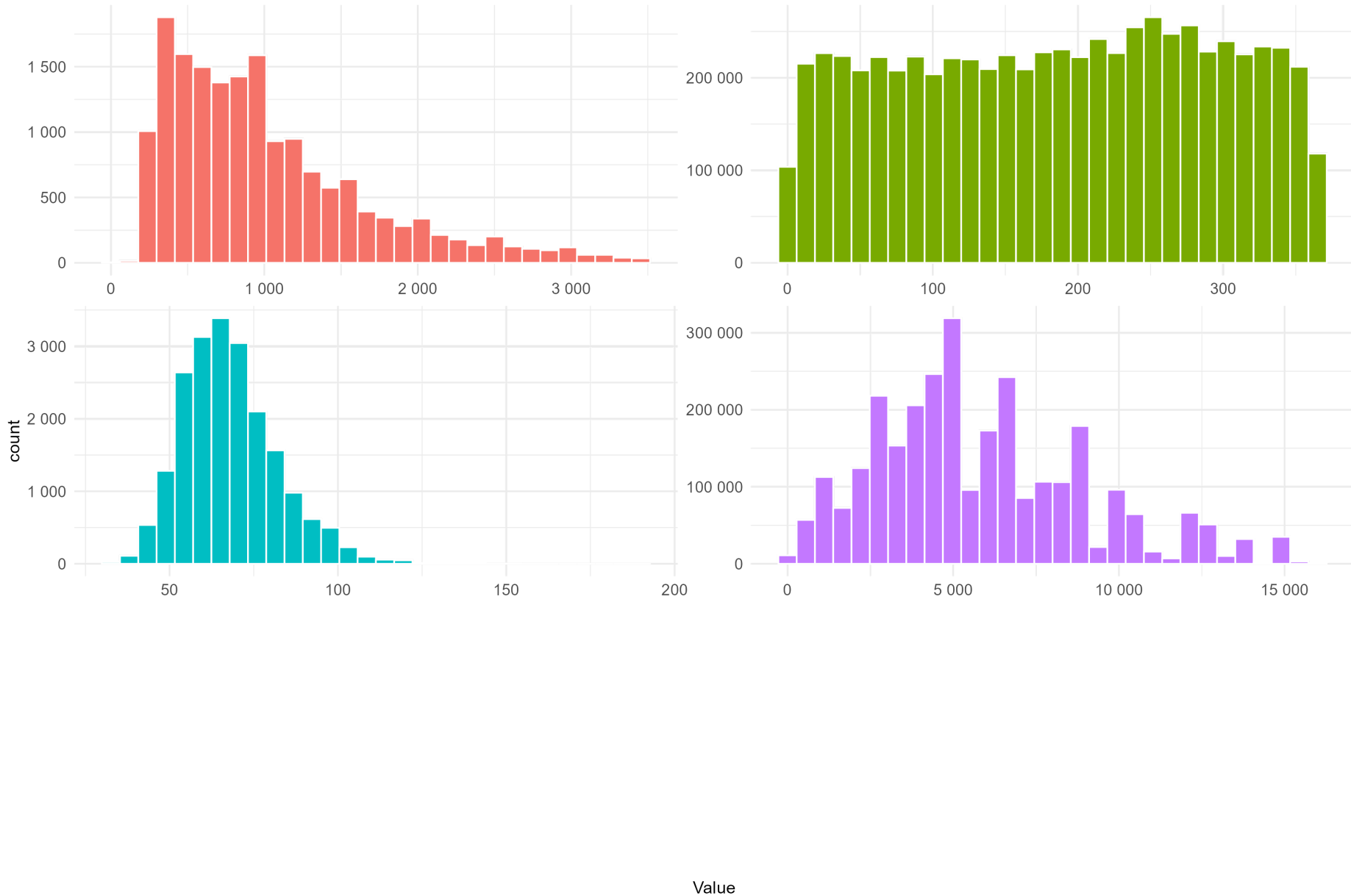
How many “peaks” does the distribution have?



Which is uniformly distributed?

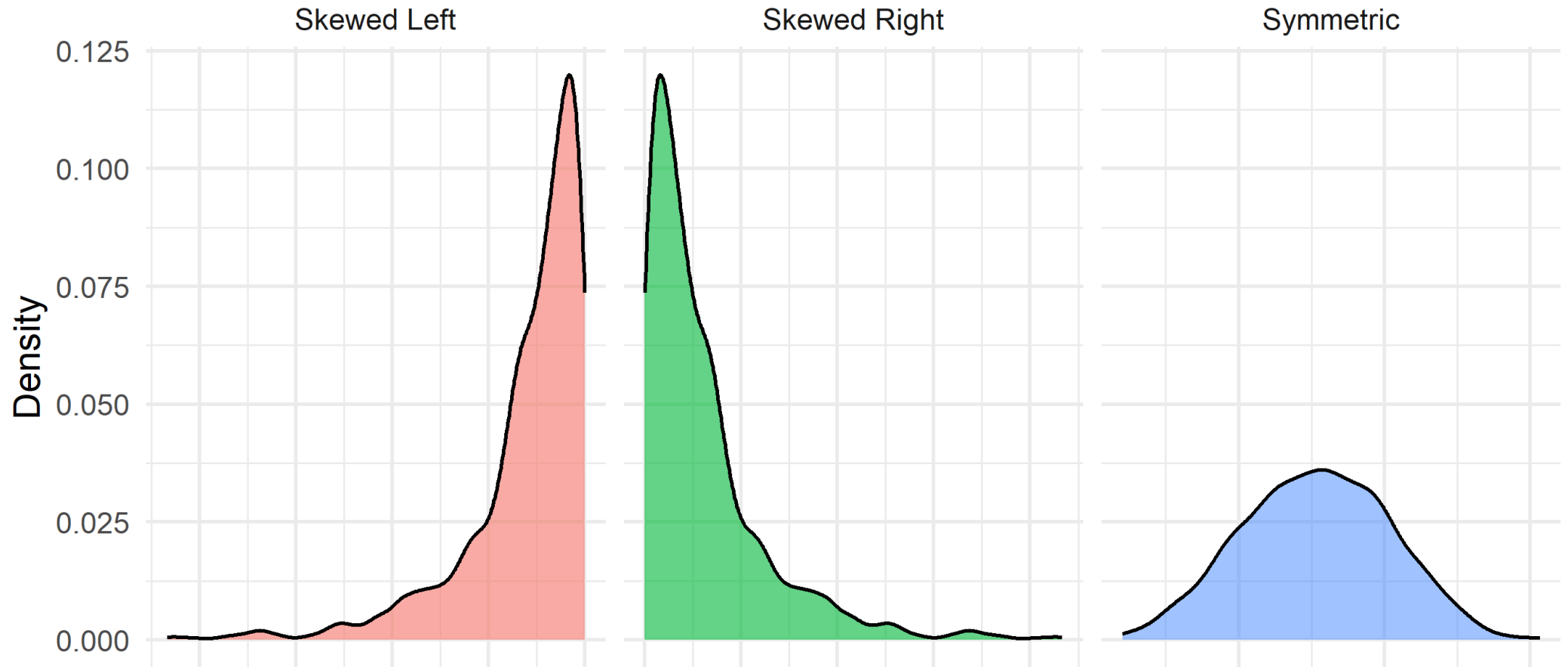
1. Weights of Adult Females
 2. Salaries in Mexico
 3. Airbnb prices in CDMX
 4. Birthdays of Classmates (day of the month)
4. Nothing makes the 7th of the month more likely than the 23rd, so every day gets roughly the same share. The other three are unimodal with a long right tail — the two price variables especially.

Which distribution is which?



Distribution Shapes: Skewness

Is the tail longer on the left or the right?



Reading skew without looking

You will not always have the picture. Two numbers you already know — the **mean** and the **median** — are enough to tell which tail is longer.

Which of the two is bigger for a right-skewed variable?



Exercise 9 — Predict, then check

You have already drawn the histogram of `weight`. **Commit to a guess first:** is its mean above or below its median? Write it down, then compute both.

Then repeat for `age`, and draw the histogram to see whether the two numbers told you the truth about the shape.

Exercise [↺ Start Over](#) [⚠ Show Solution](#) [▶ Run Code](#)

```
1 # my guess for weight: mean is ____ the median
2 # my guess for age:      mean is ____ the median
```

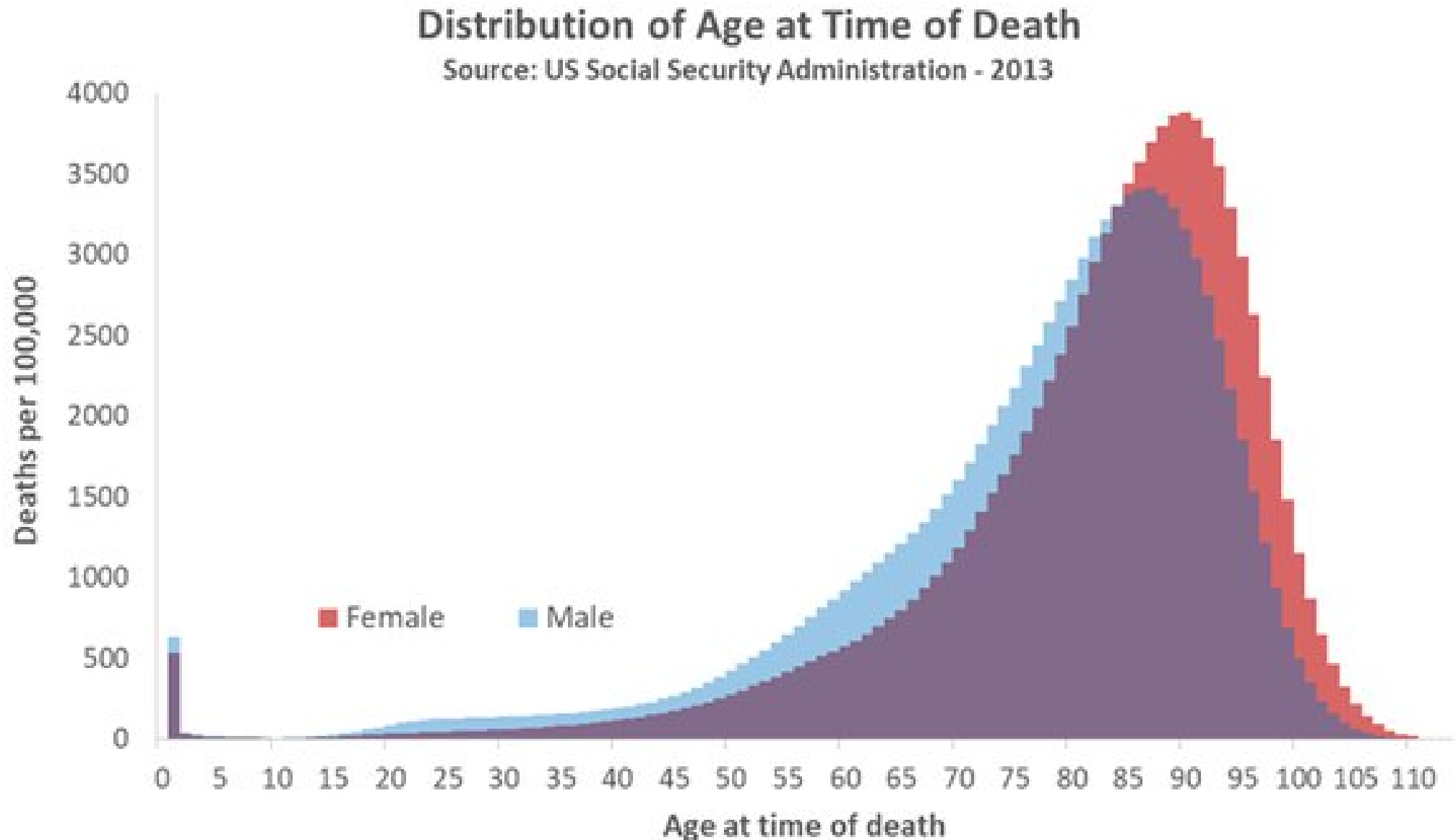
If the mean is clearly larger than the median, the distribution is...

Right-skewed — the long right tail pulls the mean up above the median.

Shape in R: the analyst's move

- The whole job of reading a distribution: **plot it, then name its shape.**
- **Modality** — count the peaks: one (unimodal), two (bimodal), many.
- **Skewness** — which tail is longer: left, right, or symmetric.
- In R: `ggplot(df, aes(x = var)) + geom_histogram()`, then eyeball it.
- That's enough to answer most business questions — do it, don't derive it.

Age at death (a left-skewed variable)



Exam Spotlight

The real questions — and how we solve them.

→ [More exam practice at the end of this deck](#)

Exam question — Fall 2025 Midterm 1

Real question from last year's Midterm 1 — **Households & Environment Survey** (each household surveyed once, 2017).
We solve this together in class.

Q3 (Distributions). *An energy company wants to understand the distribution of weekly household gasoline spending to design a fuel discount program.*

- a. **Visualize** the distribution of `gasoline_spending_weekly`.
- b. **Describe its shape** — modality and skewness.

Pick the right plot (a **histogram**), then read the **shape** straight off it: one peak or several, and which tail is longer.

Practice now (this week's pack): the histogram + shape-description problems in the practice set.

Exam question — Fall 2025 M1 (cont.)

Q5 (Boxplot interpretation). The boxplot compares **monthly electricity bills** for households that use electricity for cooling (AC/fans) vs those that do not.

- a. Which group has the **higher median** electricity bill?
- b. Which group shows **greater variability (spread)**?

Read it straight off the five-number summary: **median line** for (a), **box width / whisker length** for (b).

Practice now (this week's pack): the boxplot problems in the practice set.

Exercise 15 — The whole week, one question

You are the consultant for the weight-loss subscription start-up. Your boss asks:

“People with diabetes are our first market. Are they heavy enough to need a different product from everyone else — or is one product enough?”

Answer it with the tools of this week: **a number, a picture, and one sentence.**

Exercise

 Start Over

 Show Hint

 Run Code

```
1 # 1. Numbers: the typical weight of each group -- and how spread out each group is.
2 #   Which centre measure, and which spread measure? Justify the pair you pick.
3
4 # 2. A picture: one plot a manager can read in three seconds.
5
6 # 3. One sentence: what do you tell your boss?
```

What you must be able to do this week

- Read a **histogram** and describe the distribution's **shape** — modality (uni-/bi-/multimodal) and skewness (which tail is longer).
- Compute and interpret **range, IQR, variance, and standard deviation**, and know how each reacts to a linear change $y = bx + a$ and which one an **outlier** moves.
- Choose the right **spread measure** for the question — robust (IQR) vs sensitive to outliers (range, variance).
- Read a **boxplot's five-number summary** — median, quartiles, whiskers, and flagged outliers.

Next week (C3): from describing data to reasoning about **uncertainty** — **estimators and the Central Limit Theorem (CLT)**.

Appendix — optional (extra & theory)

Variance & Standard Deviation — why square then root?

- Why do we first take squares and then take square root?
- Can't we just do $\frac{1}{N} \sum_{i=1}^N (x_i - \mu)$?
- NO! Because $\sum_{i=1}^N (x_i - \mu) = 0$

Variance & Standard Deviation (cont.) — why not MAD?

- Why don't we just do Mean Absolute Deviation?

- $MAD = \frac{1}{N} \sum_{i=1}^N |(x_i - \mu)|$

- MAD is not differentiable at 0 :(
- Variance puts more weight on far away observations.
- It's a weighted average distance, where weights are distances themselves.

$$\sigma^2 = \sum_{i=1}^N (x_i - \mu)^2 = \frac{1}{N} \sum_{i=1}^N \underbrace{|(x_i - \mu)|}_{Weight} * \underbrace{|(x_i - \mu)|}_{Distance}$$

- It makes it very sensitive to outliers, which get a lot of weight!
- Standard deviation retains this property

Coefficient of Variation

Coefficient of Variation divides the standard deviation by the mean.

$$C.V. = \frac{\sigma}{|\mu|}$$

And sample equivalent

$$c.v. = \frac{s}{|\bar{x}|}$$

- Why?
 - It expresses standard deviation as proportion of the mean
 - Small value means variation is low compared to the mean
 - It is unit free
 - You can compare it across variables with different units/magnitudes

Coefficient of Variation: example

Example — two stocks, priced in different currencies. Which one is riskier?

Date	MXN stock	USD stock
2023-07-01	91.59	1.01
2023-07-02	96.55	1.16
2023-07-03	123.38	1.02
2023-07-04	101.06	1.07
2023-07-05	101.94	1.09
2023-07-06	125.73	0.90
2023-07-07	106.91	1.35
2023-07-08	81.02	1.23

- One trades around 100 pesos, the other around 1.2 dollars.
- Their standard deviations are *not* comparable — different units, different magnitudes.
- So which stock moves around more?

Exercise 13 — Make two incomparable things comparable

The two price series are rebuilt for you. Compute the SD of each, then the coefficient of variation of each, and say which stock is more variable — and why the two statistics disagree about the answer.

Then do the same trick on `Health_data`: `weight` (kg) and `age` (years) have almost identical SDs. Is age therefore just as variable as weight?

Exercise

 Start Over

 Show Hint

 Run Code

```
1 set.seed(123)
2 mxn <- round(rnorm(20, mean = 100, sd = 15), 2)
3 usd <- round(rnorm(20, mean = 1.2, sd = 0.18), 2)
```

Coefficient of Variation: the answer

- **Standard deviation:**
 - USD: 0.149
 - MXN: 14.59
- **Coefficient of variation:**
 - USD: 0.12
 - MXN: 0.14

The SD ratio is an artefact of the currency. The CV is unit-free, so it is the one you can compare.

Coefficient of Variation: transformations

So more generally, if $y_i = bx_i$, then

$$C.V._y = \frac{\sigma_y}{|\mu_y|} = \frac{|b|\sigma_x}{|b\mu_x|} = C.V._x$$

What if $y_i = bx_i + a$?

Then

$$C.V._y = \frac{\sigma_y}{|\mu_y|} = \frac{|b|\sigma_x}{|b\mu_x + a|} \neq C.V._x$$

Standard Deviation (cont.) – $Y = X + 40$

$$\mu_Y = E(Y) = \frac{\sum_{i=1}^N (y_i + 40)}{N} = \frac{\sum_{i=1}^N (x_i + 40)}{N} = \frac{\sum_{i=1}^N x_i + N * 40}{N} = E(X) + 40 = \mu_X + 40$$

$$\sigma_Y = \sqrt{Var(Y)} = \sqrt{\frac{\sum_i (y_i - \mu_y)^2}{N}} = \sqrt{\frac{\sum_i ((x_i + 40) - (\mu_x + 40))^2}{N}} = \sqrt{\frac{\sum_i (x_i - \mu_x)^2}{N}} = \sqrt{Var(X)} = \sigma_X$$

$$E(X + c) = E(X) + c \quad \text{and} \quad Var(X + c) = Var(X)$$

Standard Deviation (cont.) – $Y = 3X$

$$\mu_Y = E(Y) = \frac{\sum_i y_i}{N} = \frac{\sum_i 3x_i}{N} = 3 \frac{\sum_i x_i}{N} = 3E(X) = 3\mu_X$$

$$\sigma_Y = \sqrt{Var(Y)} = \sqrt{\frac{\sum_i (y_i - \mu_y)^2}{N}} = \sqrt{\frac{\sum_i (3x_i - 3\mu_x)^2}{N}} = \sqrt{3^2 \frac{\sum_i (x_i - \mu_x)^2}{N}} = \sqrt{3^2 Var(X)} = 3\sigma_X$$

$$E(cX) = cE(X) \quad \text{and} \quad Var(cX) = c^2 Var(X)$$

Exam question — Fall 2025 M1 (associations)

Q4 (Associations). *A transport consultant needs to know what drives gasoline spending. Which is more strongly associated with **weekly gasoline spending** — the **number of vehicles** or the **household size**? Choose an appropriate metric, compute it, and interpret.*

Would the conclusion change if spending were in USD instead of MXN (1 USD = 17 MXN)?

This is a **correlation vs covariance** question: pick the metric, compute both associations, compare.

Behind every histogram there is a table

The exam asks for the **table**, not the picture: interval, midpoint, absolute frequency n_i , relative frequency p_i , and cumulative frequency P_i .

Build it before you see ours.



Exercise 18 — Build the frequency table

Cut `weight` into 5 equal intervals and produce the five columns. Then answer: what share of people weigh less than 94.55 kg?

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

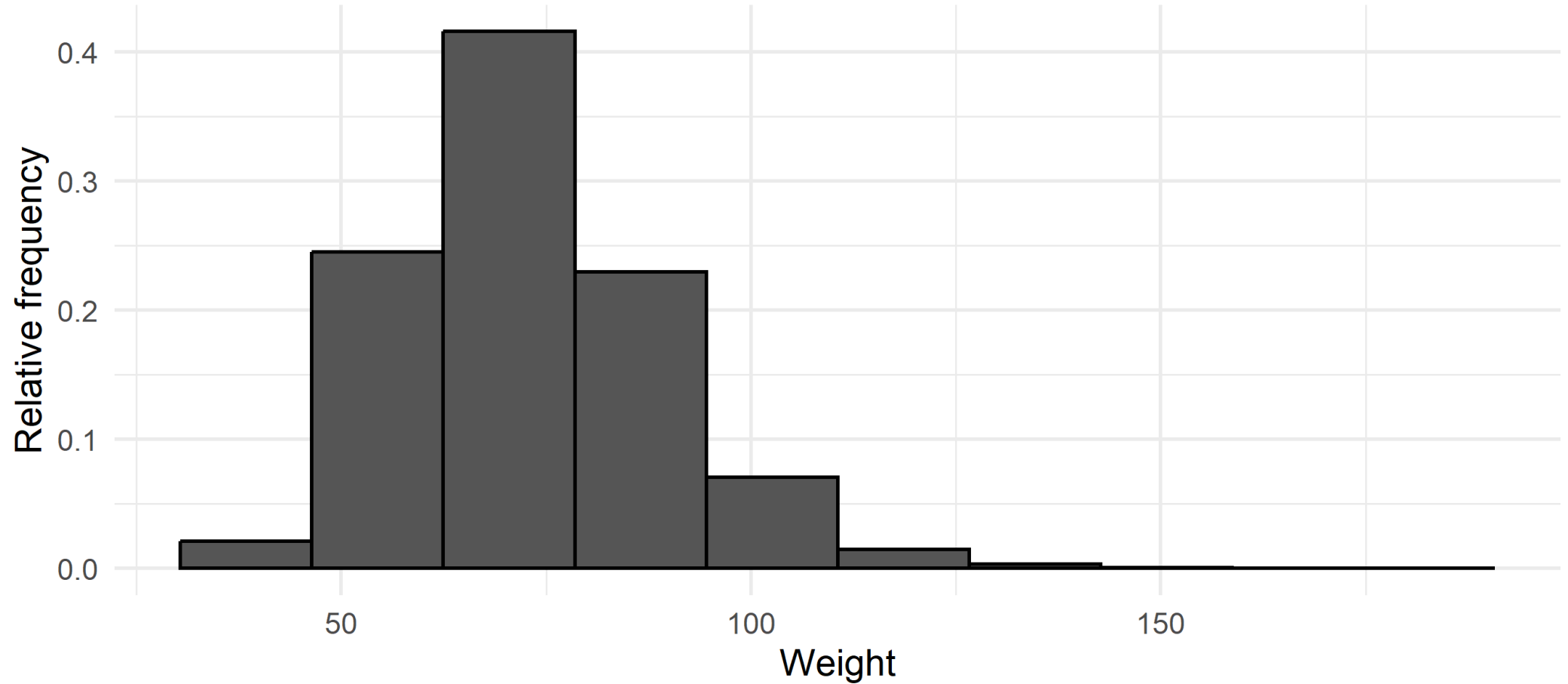
```
1 h <- hist(Health_data$weight, breaks = 5, plot = FALSE)
2 h$breaks
3 h$counts
4
5 # build: interval, midpoint, n_i, p_i, P_i
```

Histogram with 5 classes: the table

Interval	Midpoint	n_i	p_i	P_i
30.37 - 62.46	46.42	10068	0.266	0.266
62.46 - 94.55	78.50	24430	0.645	0.911
94.55 - 126.63	110.59	3206	0.085	0.996
126.63 - 158.72	142.68	143	0.004	1.000
158.72 - 190.81	174.76	11	0.000	1.000

Histogram with 10 classes

More, narrower bins reveal more detail.



Histogram with 10 classes: the table

Interval	Midpoint	n _i	p _i	P _i
30.37 - 46.42	38.40	796	0.021	0.021
46.42 - 62.46	54.44	9272	0.245	0.266
62.46 - 78.5	70.48	15742	0.416	0.682
78.5 - 94.55	86.53	8688	0.229	0.911
94.55 - 110.59	102.57	2661	0.070	0.982
110.59 - 126.63	118.61	545	0.014	0.996
126.63 - 142.68	134.66	126	0.003	0.999
142.68 - 158.72	150.70	17	0.000	1.000
158.72 - 174.76	166.74	8	0.000	1.000
174.76 - 190.81	182.79	3	0.000	1.000

Histogram with 100 classes: the table

Interval	Midpoint	n_i	p_i	P_i
30.37 - 31.98	31.18	8	0.000	0.000
31.98 - 33.58	32.78	11	0.000	0.001
33.58 - 35.19	34.39	7	0.000	0.001
35.19 - 36.79	35.99	16	0.000	0.001
36.79 - 38.4	37.59	24	0.001	0.002
38.4 - 40	39.20	24	0.001	0.002
40 - 41.6	40.80	114	0.003	0.005
41.6 - 43.21	42.41	126	0.003	0.009

First 8 of 100 intervals shown.

Probability Density Function (PDF)

- The **pdf** describes the probability distribution of a **continuous** random variable.
- It does **not** give the probability at a single value (always 0 for a continuous variable)
- It shows in which intervals the variable most often appears
- Used to compute the probability of the variable falling in an interval
- **Area under it always adds up to 1**

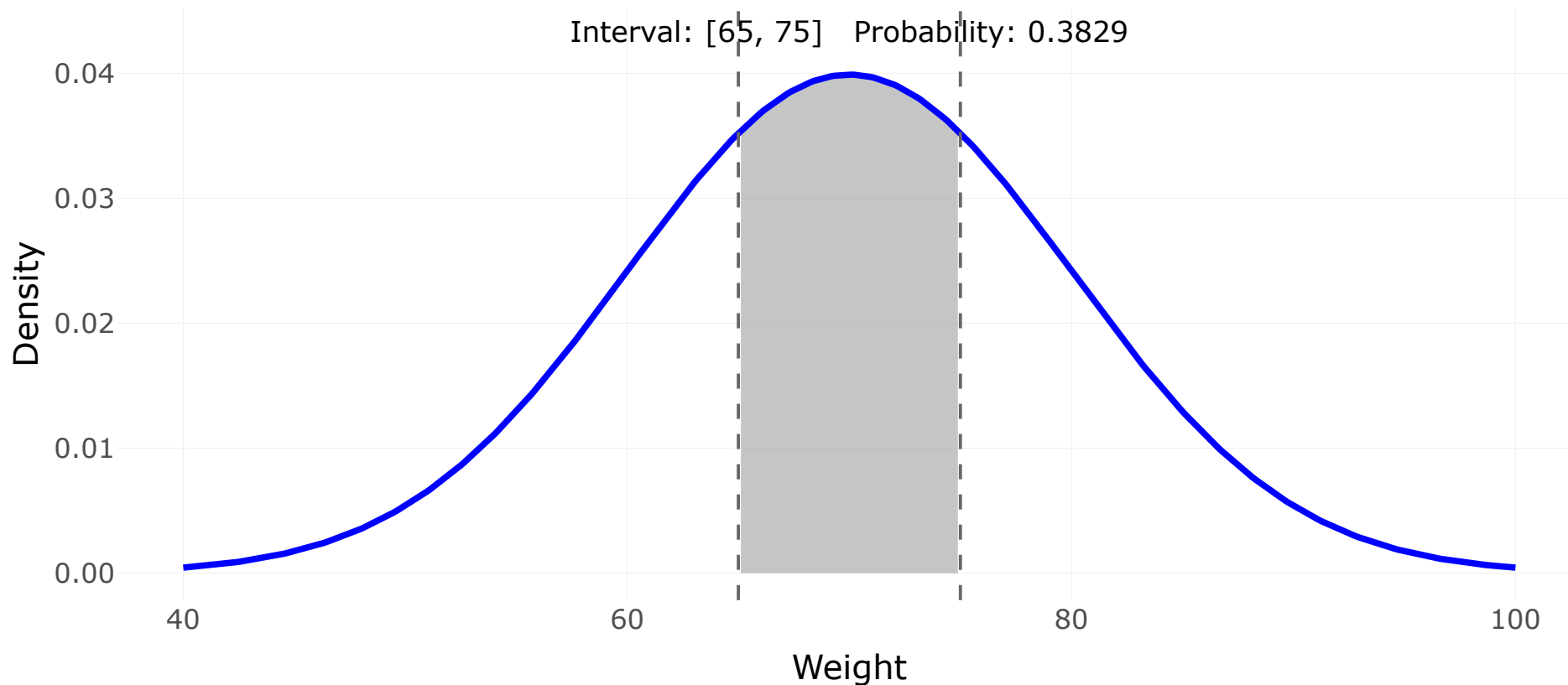
Example: X = weight of adults in Mexico. The PDF describes how likely a weight is to fall in a range (e.g. between 58 and 60 kg).

PDF: computing a probability over a range

To get the probability that X falls in $[a, b]$, integrate the PDF from a to b :

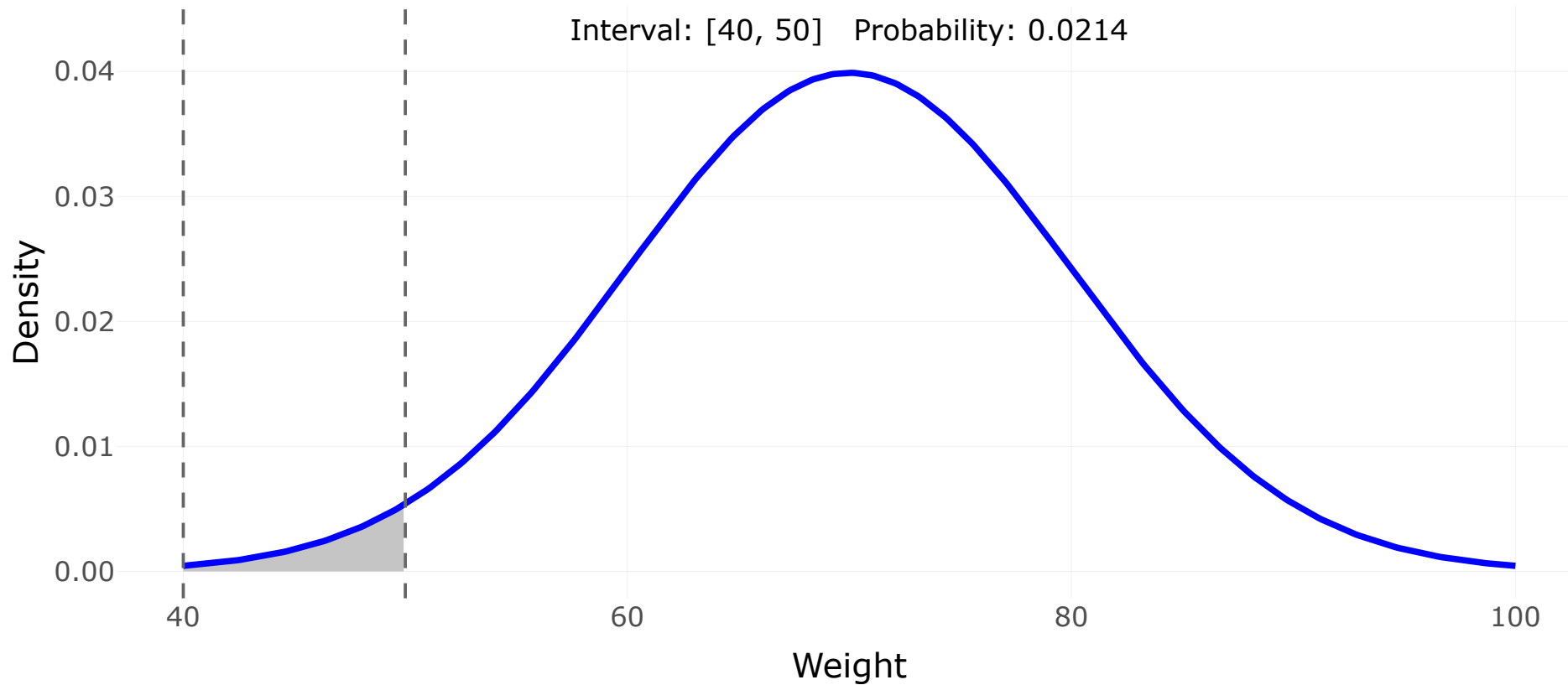
$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

What share of the population weighs between **65 and 75 kg**?



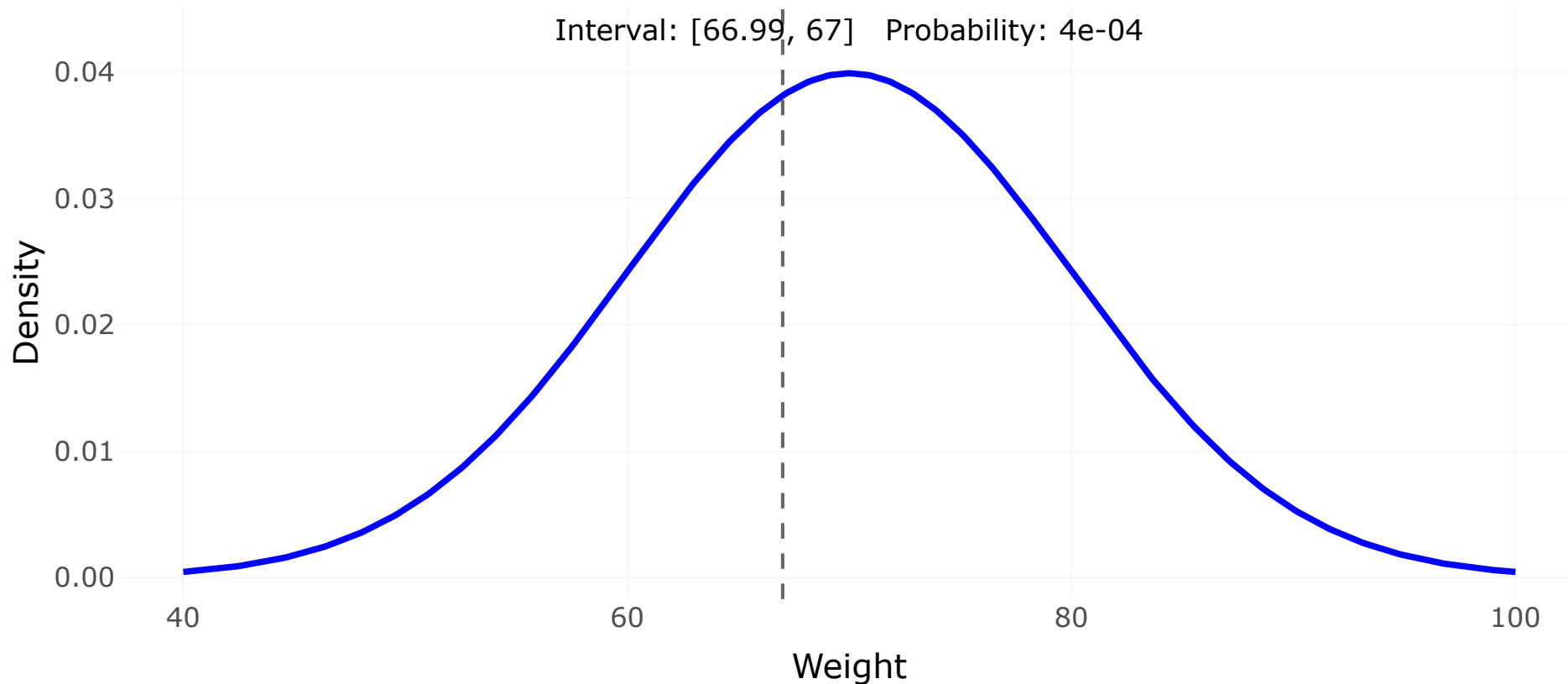
PDF: a wider range

What share of the population weighs between 40 and 50 kg?



PDF: a razor-thin range

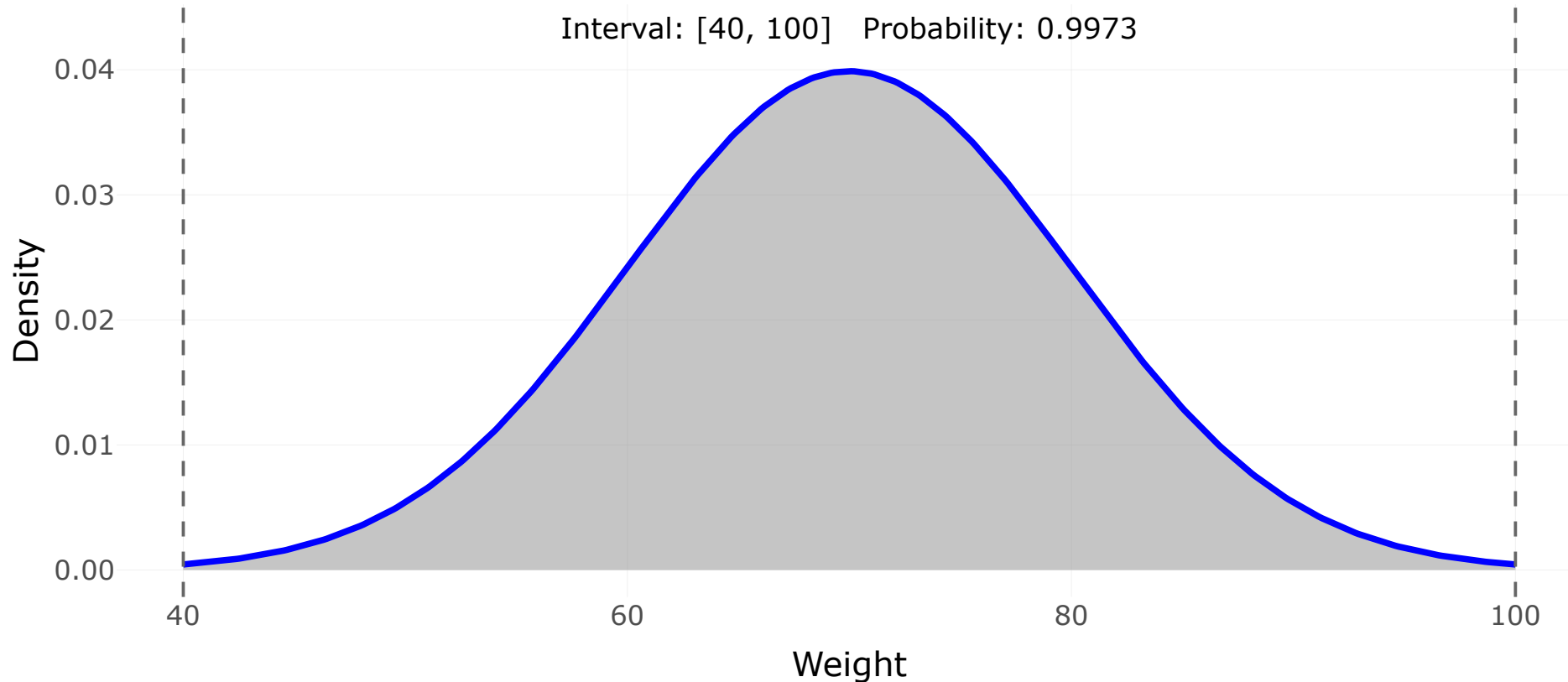
What share of the population weighs between 66.99 and 67 kg?



A single point has ~0 probability — that's why the pdf gives **densities**, not probabilities.

PDF: the whole range integrates to 1

What share of the population weighs between 40 and 100 kg? (almost everyone)



- The area under the **entire** curve is 1.

From PDF to CDF

What if we want the proportion of people who weigh **50 kg or less**?

We need the *cumulative* picture — enter the CDF.

Cumulative Distribution Function (CDF)

The **Cumulative Distribution Function** (CDF) gives the probability that X takes a value **less than or equal to** a specific value.

For a continuous X with PDF $f(x)$, the CDF is:

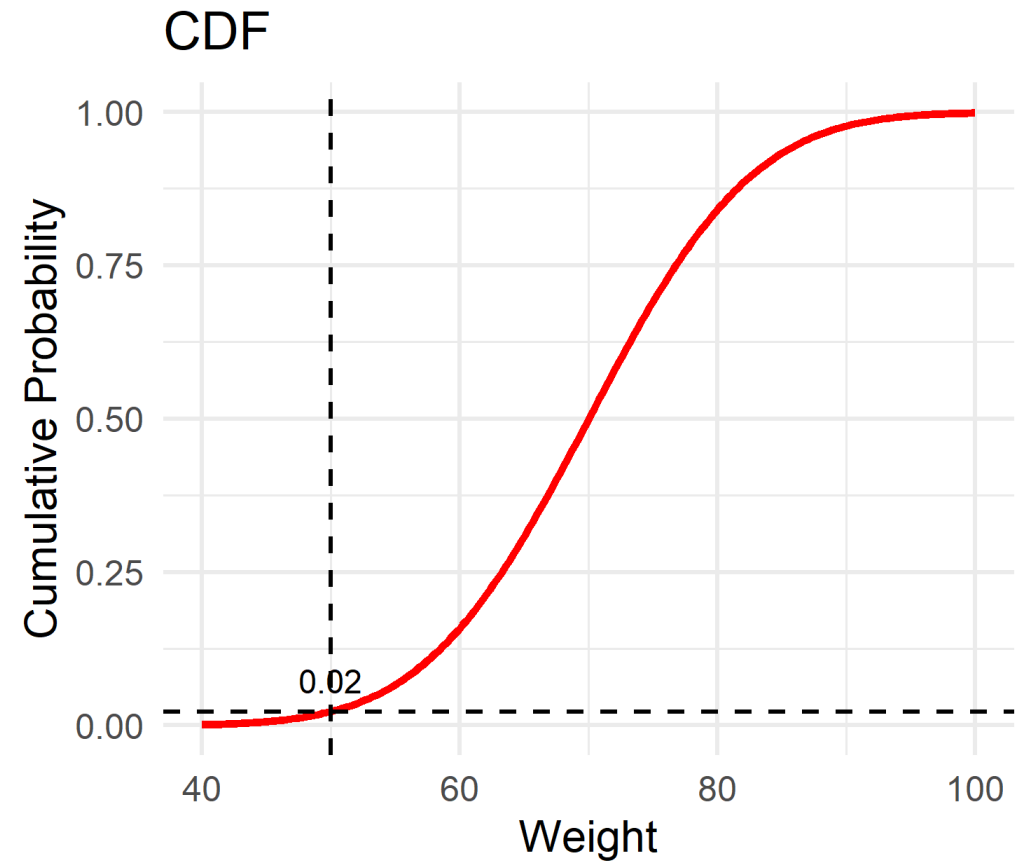
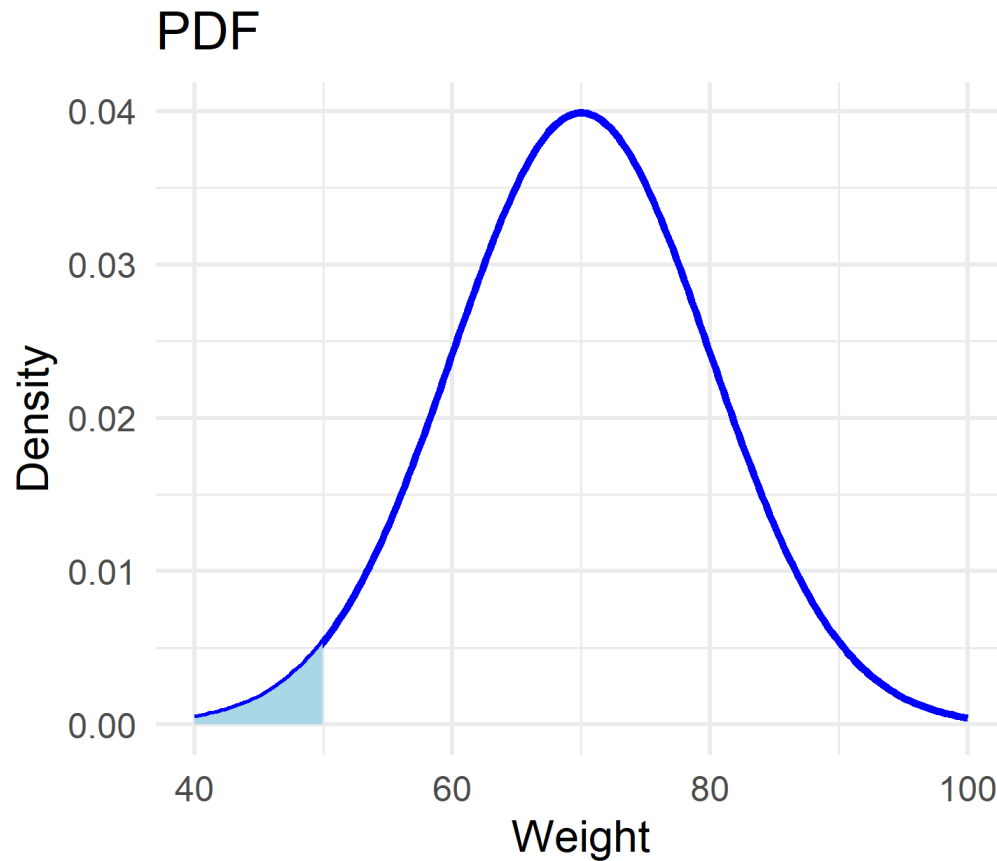
$$F(x) = \int_{-\infty}^x f(t) dt = P(X \leq x)$$

Characteristics:

- Starts at 0 (as $x \rightarrow -\infty$) and approaches 1 (as $x \rightarrow +\infty$)
- Non-decreasing and right-continuous

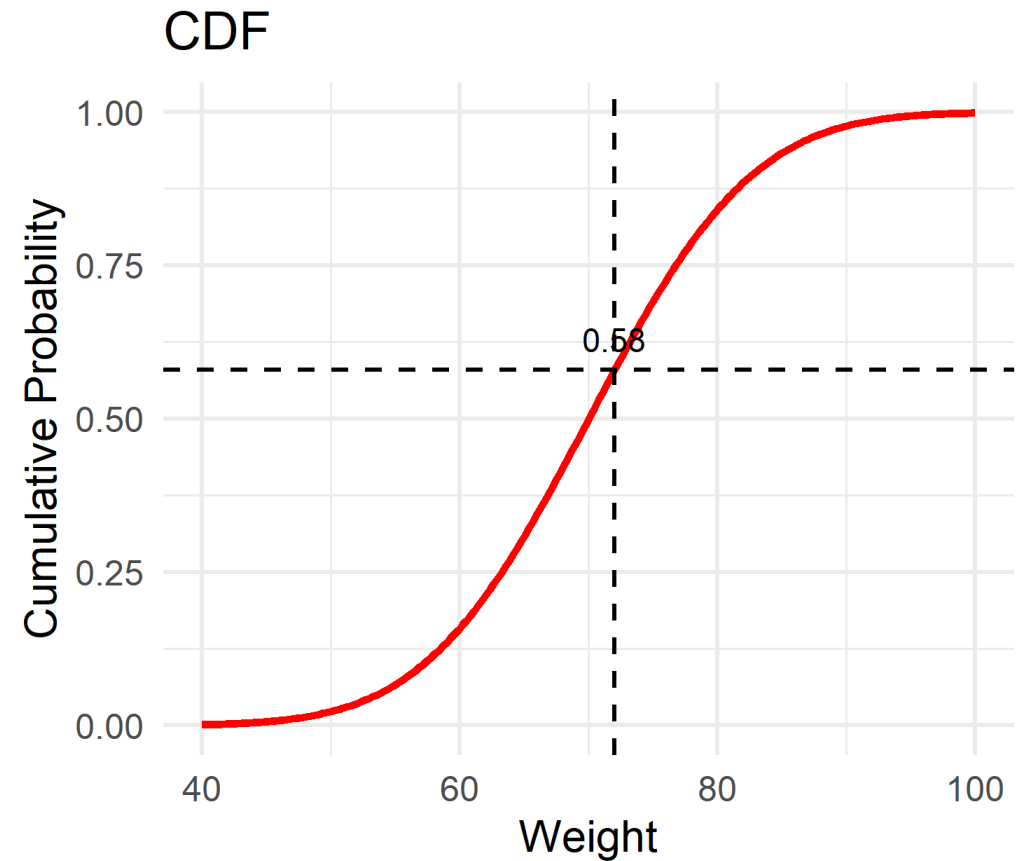
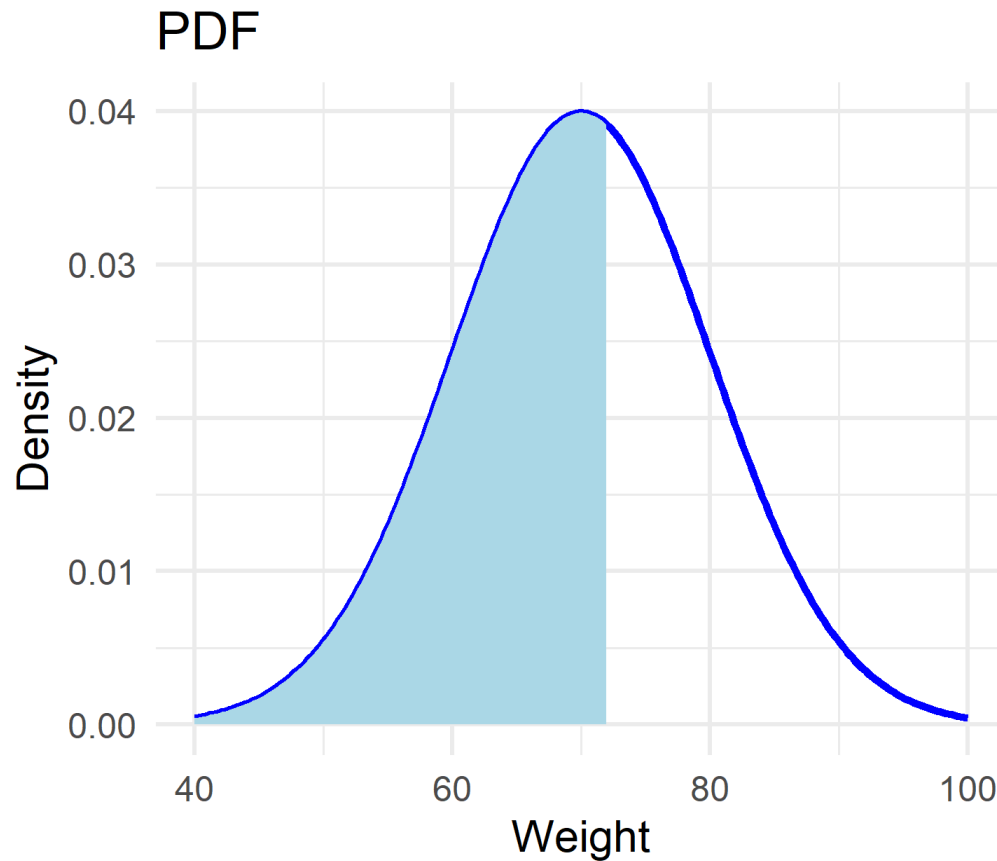
CDF Example 1: weight in the population

$$F(50) = P(X \leq 50) = 0.02$$



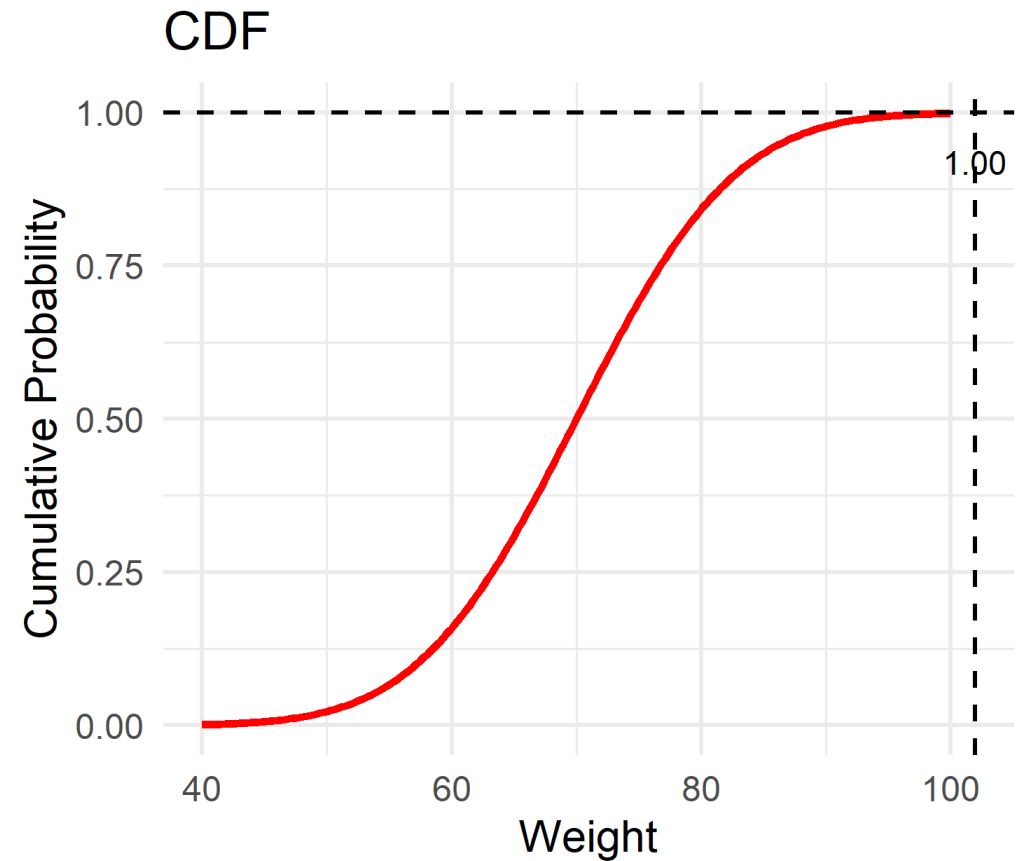
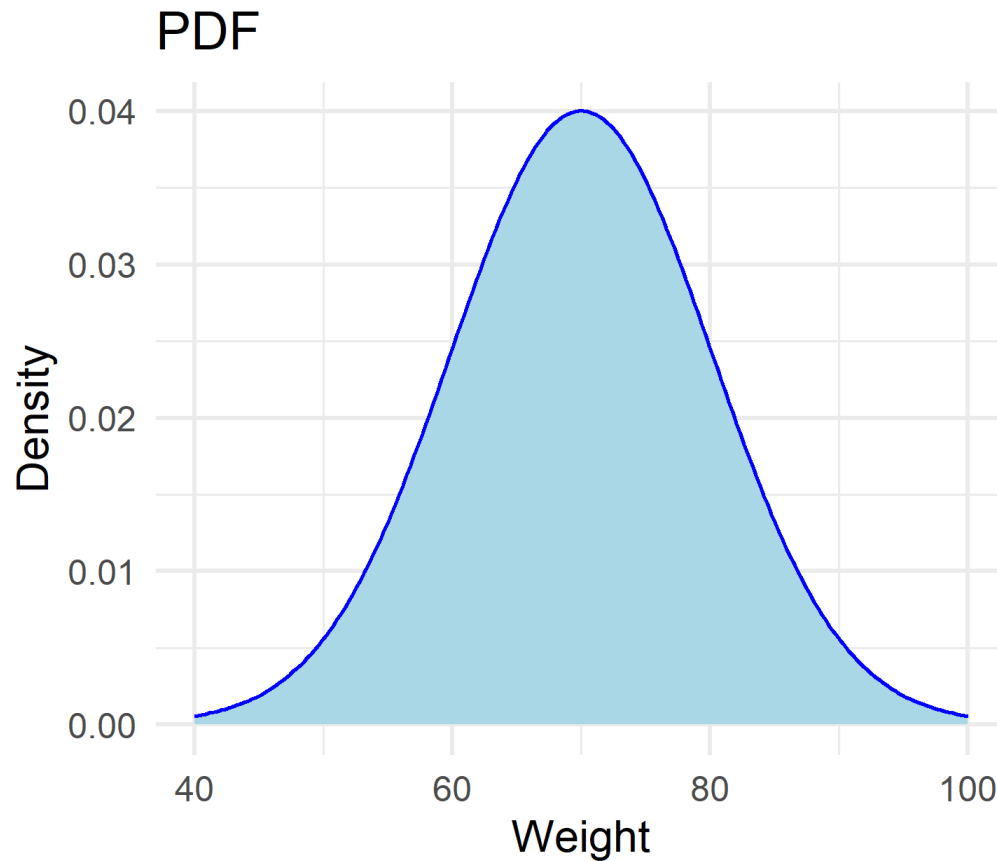
CDF Example 2: weight in the population

$$F(72) = P(X \leq 72) = 0.58$$



CDF Example 3: weight in the population

$$F(102) = P(X \leq 102) = 0.99$$

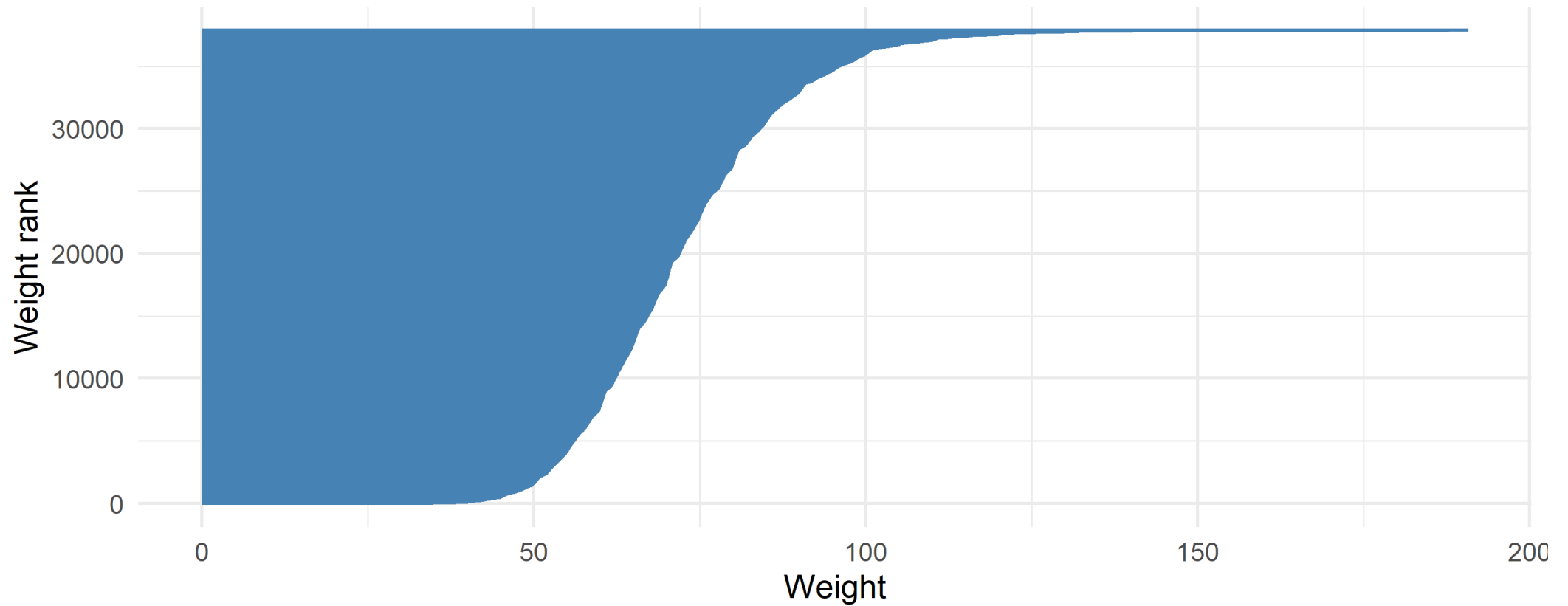


Never integrate a CDF! (it is already the integral of the pdf)

Empirical CDF

What if we only have a **sample** and don't know the true pdf? Line up everyone's weight and sort it.

Sorted by weight



Empirical CDF (cont.)

$$ECDF(x) = \frac{\sum I(w_i \leq x)}{N} = \frac{\text{Number of people with weight} \leq x}{N}$$

$I(w_i \leq x) = 1$ if person i 's weight is $\leq x$ (*indicator function*); N = sample size.

Evaluate that at every x and you get a curve. What does it look like?

Empirical CDF: finding percentiles

- Say we want the 25th percentile (1st quartile), call it v_{25}
- By definition $P(X \leq v_{25}) = 25\%$
- So the 1st quartile is the value where the CDF equals 25%
- More generally, percentile Z : $P(X \leq v_Z) = Z\%$



Exercise 10 — Read a percentile off your own ECDF

Build the ECDF of `weight` yourself. Draw the horizontal line at 0.25, drop a vertical line where the curve crosses it, and **read the 25th percentile off your picture** — then check the number with `quantile()`.

Then use the curve in the other direction: what share of people weigh 100 kg or less?

Exercise

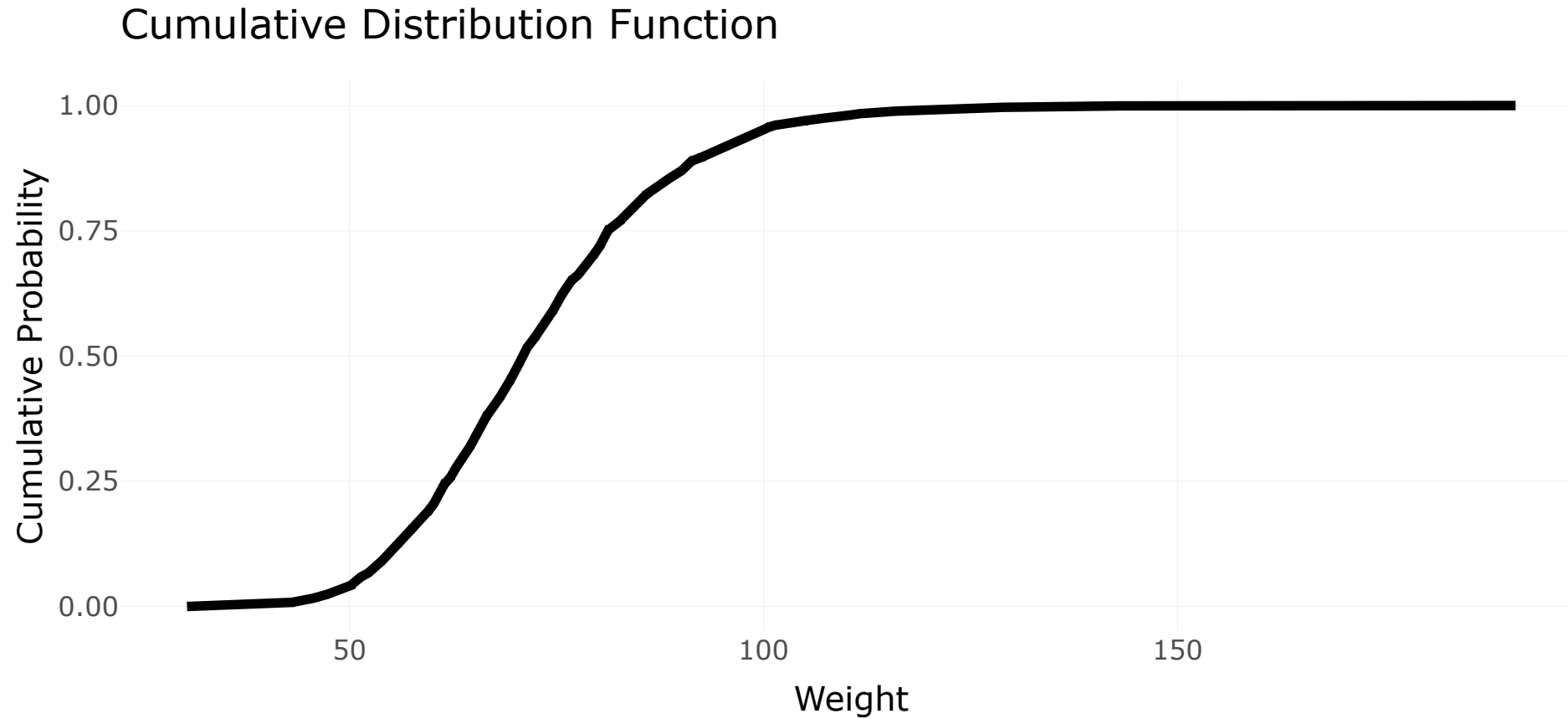
↺ Start Over

💡 Show Hint

▶ Run Code

```
1 e <- ecdf(Health_data$weight)
2 # 1. plot it, 2. mark 0.25, 3. read off the weight, 4. check it
```

Empirical CDF: the curve

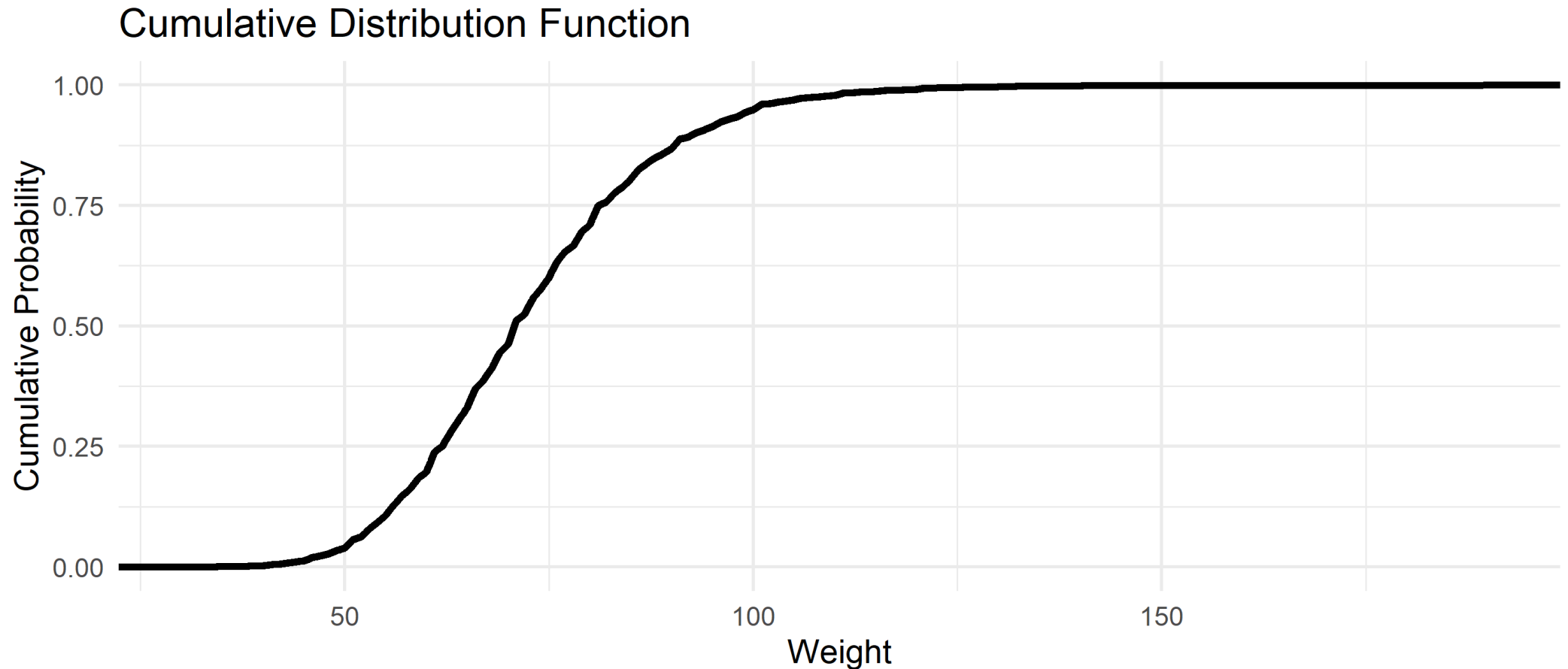


Empirical CDF: reading it

- Share weighing ≤ 50 kg: $P(\text{weight} \leq 50) = ECDF(50)$.
- Share weighing more than 100 kg: $1 - ECDF(100)$.

Empirical CDF: finding percentiles (cont.)

Read the percentile off the CDF:



Appendix — scenario & population vs sample

This week: distributions, fast and hands-on

- This is **review** — you've met distributions before. We move quickly and **do it in R**.
- What a business analyst needs to *read* a distribution: **histograms** and **shape** (modality, skewness).
- We also go a bit deeper on the theory: the **PDF and CDF** sections.
- Behind on the underlying stats? Study those sections; if you're solid, skim them.
- Practice as we go with the exercise slides embedded in this deck.

Summary

- In the last class:
 - We discussed the organization of the course
 - We overviewed forecasting methods
 - We learned about methods of qualitative forecasting
 - *Reference:* Forecasting Methods and Applications, chapter 1
- This set of classes:
 - We will start learning about exploratory analysis preparing the forecast
 - We will learn to translate business problems to statistical problems
 - We will learn about various **data types**
 - We will learn how to **summarize data and observed patterns**
 - *Reference:* Forecasting Methods and Applications, chapter 2.1-2.4

Why do we care?

- To use scientific methods we need to set business problem as a statistical problem
- Define things that we want to find to identify appropriate methods

Specific Scenario

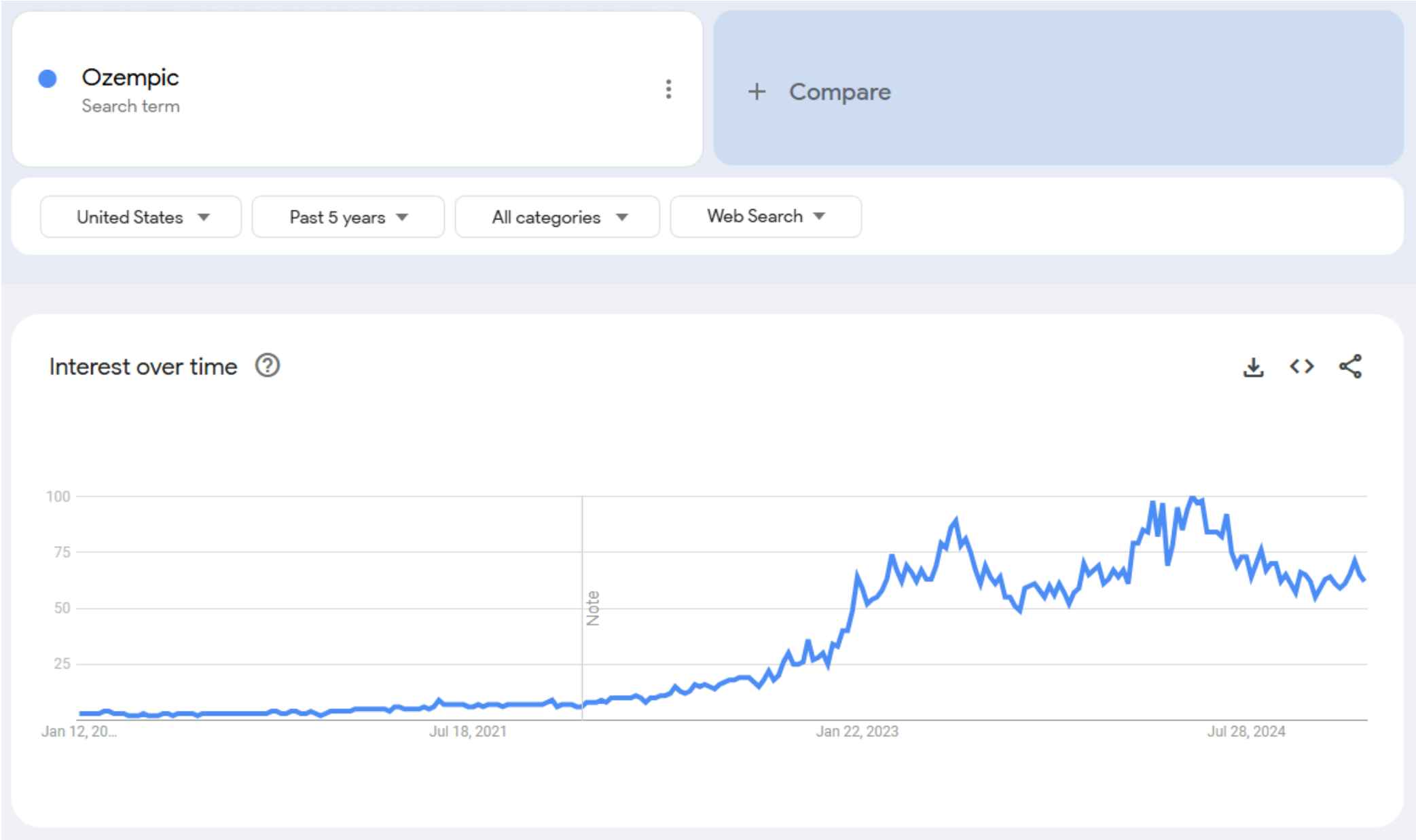
The Opportunity

- Online pharmacies are rapidly growing, offering convenience and accessibility to customers.
 - Example in Mexico: *Choiz*, which provides prescription services and drug subscriptions.

The Trend

- A new generation of highly effective anti-diabetes medications, like *Ozempic*, is gaining popularity for dual benefits: managing diabetes and aiding weight loss.

Search Interest in Ozempic



Your Role

- You have been hired as a consultant for a start-up aiming to launch a subscription service for these medications in Mexico.



The Challenge

- Your boss needs a detailed exploratory market analysis to forecast sales and identify key customer segments

Exam practice

These problems mirror this week's exercise set (D6–D14) and the exam. Each one gives you the data — **run the commands, look at what comes back, then decide**. Press **Solution** once you have your own answer.

[← Back to the Exam Spotlight](#)



Exam practice 1 — Exploring the cps dataset

The `cps` dataset (loaded for you) is a labor force survey with individuals' demographic characteristics, employment status, wages, and education. Variables of interest: [D6]

- `lfstatus`: labor force status (employed, unemployed, not in labor force)
- `marstatus`: marital status (married, single, divorced, widowed)
- `race`: race category (e.g., White, Black, Other)
- `wagehr`: hourly wage (in dollars, only observed for individuals paid by the hour)
- `ownchild`: number of own children in the household

(a) Provide a frequency table of `lfstatus`. **(b)** Draw a bar chart of the proportions of `lfstatus`. Interpret the results.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Your code here
```



Exam practice 1 — cps dataset (cont.)

(c) Create a relative conditional frequency table of `marstatus` and `race`. (d) What is the difference in the probability of being married between Black and White individuals? [D6]

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Your code here
```



Exam practice 1 — cps dataset (cont. 2)

(e) Calculate the difference in the average hourly wage (`wagehr`) between married and not-married individuals. (f) Restrict to individuals aged 30–59: what are the mean and standard deviation of `ownchild`? [D6]

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Your code here
```



Exam practice 1 — cps dataset (cont. 3)

(g) The variable `wagehr` has the following summary statistics: [D6]

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
1.01	12.78	16.41	18.60	22.00	90.00	2174

i. Is the distribution symmetric, left-skewed, or right-skewed? ii. If one more worker with wage = 16.41 is added, how would the sample mean and median change?

Get the evidence out of R before you answer — then check whether the table alone would have been enough to reach the same conclusion.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 summary(cps$wagehr)
2
3 # i. draw it, and compare the mean with the median
4 # ii. add one worker at 16.41 and recompute both
```

i. **Right-skewed.** The mean (18.60) exceeds the median (16.41), and the distance from the median to the max (73.59) is much larger than from the min to the median (15.40). This is typical of wage distributions.

ii. The new wage equals the median, so the **median** stays the same or barely changes. The **mean** decreases slightly because $16.41 < 18.60$ (the current mean), pulling the mean down.



Exam practice 1 — cps dataset (cont. 4)

(g), **continued**. Now compute: **iii.** the sample interquartile range (IQR) of `wagehr`; **iv.** the values of the lower and upper whiskers in the box plot of `wagehr`. [D6]

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Your code here
```



Exam practice 2 — Website arrivals

Use the `website` dataset (loaded for you), a time-series dataset containing visitor arrivals. [D7]

(a) Draw a histogram of `arrival1`. How would you describe the modality of this distribution?

Exercise

↺ Start Over

⚠ Show Solution

▶ Run Code

```
1 # Your code here
```



Exam practice 3 — Aguas frescas (by hand)

A small business sells aguas frescas at a weekend tianguis in Mexico City. Each cup is sold for 40 pesos. Over 16 Sundays, the number of cups sold per day has a sample mean of $\bar{x} = 120$ and a sample standard deviation of $s_x = 20$. [D8]

- (a) What are the sample mean and standard deviation of **revenues** in pesos? (*Revenue = price $\times$ quantity.*)
- (b) The stall rental fee is 2,000 pesos per Sunday, and it costs 5 pesos to produce each cup. What are the sample mean and standard deviation of **profits**? (*Profit = Revenue – Fixed Cost – Variable Cost.*)

Use R as a calculator — apply the linear-transformation rules for mean and SD:

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 xbar <- 120
2 s_x  <- 20
3 # Your code here
```



Exam practice 4 — Inflation panel

Use the `inflation` dataset (loaded for you): annual inflation rates for 45 countries, 2010–2019. Variables: `country` (3-letter code), `year`, `inflation` (percentage points). Briefly interpret each result. [D9]

- (a) How many observations are associated with deflation (negative inflation rates)?
- (b) How many countries experienced deflation at least once during this period?
- (c) Which countries have the lowest and highest *average inflation* over the 10-year period?
- (d) Which countries have the lowest and highest *standard deviation of inflation*?

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Your code here
```



Exam practice 4 — Inflation panel (cont.)

(e) Draw a time-series plot for inflation in the United States (USA). (f) Draw a time-series plot showing inflation for USA, Canada (CAN), and Mexico (MEX) on the same plot — make sure each country's line is distinguishable. [D9]

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Your code here
```



Exam practice 5 — Wages by race

Use the `cps` dataset again. [D10]

(a) Plot the average hourly wage (`wagehr`) by race. What does it show? Report the group means and compare across races. Interpret carefully: are there differences in wages across demographic groups? How might economists explain them?

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Your code here
```



Exam practice 6 — eBay auctions

Use the `auctions` dataset (loaded for you): 684 eBay auctions of iPod Mini devices in 2006. Key variables: `finalprice` (winning bid), `bidders` (number of unique bidders), and condition dummies (`new`, `used`, `refurb`). [D11]

- (a) Focus only on the subsample of 624 new and used devices (drop those with `refurb` = 1).
- (b) Draw a scatter plot of `finalprice` versus `bidders`.
- (c) Compute the sample correlation between `finalprice` and `bidders`. Why is this value interesting in the context of auctions?
- (d) Based on the plot in (b), does `finalprice` always increase with more bidders, or is the relationship more nuanced? Explain.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Your code here
```



Exam practice 7 — Units of measurement

For a sample of Mexican manufacturing firms we observe x = electricity purchased (in kilowatt-hours, kWh) and y = monthly revenues (in thousands of pesos). [D12]

Build such a sample and watch what each statistic does when you change the units of y from **thousands of pesos** to **pesos**. The statistic whose value does not move is the one with no units.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 set.seed(1)
2 x <- round(rlnorm(200, 9, 0.4))           # electricity, kWh
3 y <- round(0.004 * x + rnorm(200, 0, 5), 1) # revenue, thousands of pesos
4
5 # mean(x), var(y), cov(x, y), cor(x, y) -- then redo with y in pesos (y * 1000)
```

Now answer in words. What are the units of:

(a) the sample average of x ?

kWh — the mean has the same units as the variable.

(b) the variance of y ?

(thousands of pesos)² — variance squares the deviations, so it squares the units.

(c) the covariance of x and y ?

kWh × thousands of pesos — the product of the two variables' units.

(d) the correlation of x and y ?

Unitless — the units cancel in the ratio $s_{xy} / (s_x \cdot s_y)$.



Exam practice 7 — Units of measurement (cont.)

Suppose the sample summary statistics for salary (y , in thousands of pesos) and education (x , in number of years) are: $\bar{x} = 200$, $s_x = 30$, $\bar{y} = 17$, $s_y = 3$, and $s_{xy} = 50$. [D12]

(a) What is the sample correlation between salary and education?

(b) What would the sample covariance be if salary were instead expressed in pesos (not thousands)?

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 s_x <- 30
2 s_y <- 3
3 s_xy <- 50
4 # Your code here
```



Exam practice 8 — Portfolio variance

Consider monthly returns of two Mexican stocks, A and B, listed on the *Bolsa Mexicana de Valores* (BMV). Suppose $\text{Var}(x) = 0.006$ and $\text{Var}(y) = 0.008$. [D13]

(a) What condition must the correlation $\text{Corr}(x, y)$ satisfy for the variance of the portfolio return $\frac{1}{2}(x + y)$ to be less than or equal to 0.007?

Work it out numerically — find the covariance bound, translate it into a correlation bound, and check the worst case $\rho = 1$:

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 var_x <- 0.006
2 var_y <- 0.008
3 # Your code here
```

Exam practice 9 — Hours, wages & education

Use the `cps` dataset. Key variables for this exercise: `hrs1astwk` (hours worked last week), `wagehr` (hourly wage in US dollars), `educ` (years of education), `gender` (“Male” or “Female”). [D14]

- (a) Calculate the average number of hours worked last week (`hrs1astwk`) by gender. What differences do you observe? How might labor market factors (e.g., childcare responsibilities, occupational segregation) explain these patterns?
- (b) Calculate the covariance between wages (`wagehr`) and education (`educ`). Does its sign suggest a positive or negative relationship?

Exercise ↺ Start Over 💡 Show Hint

▶ Run Code

```
1 # Your code here
```

Exam practice 9 — Currency conversion (cont.)

(c) Suppose wages are converted into Mexican pesos (assume $1 \text{ USD} = 20 \text{ MXN}$). [D14]

- i. What happens to the covariance between wages and education when wages are rescaled?
- ii. What happens to the correlation?

Verify both numerically by rescaling `wagehr` yourself, then interpret: why does correlation remain unchanged even though covariance changes?

Exercise

 Start Over

 Show Hint

 Run Code

```
1 # Your code here
```